VISION AI

Invoice System — API Testing Document

Bilingual (English / Japanese) Multi-Currency Invoicing Web Application

Backend REST API · Spring Boot (Java 17+) · Cloud MongoDB · 10 Business Modules

Document	Invoice System — API Testing Document
Version	1.0
Status	For QA execution
Product	Vision AI Enterprise Invoice & Billing System — Invoice System module
Prepared By	QA (generated from source documentation — see §0.1)
Reviewed By	To be confirmed with development / QA lead
Date	10 August 2026
Classification	Confidential — Vision AI
Purpose	Provide the QA team with a complete, traceable API-testing reference for the Invoice System: API understanding, functional/negative/boundary validation, integration testing, and regression testing.
Scope	All API-relevant behaviour of the Invoice System's 10 business modules, to the extent that behaviour is described in the source functional & test documentation (v2.7). No endpoint, field, or business rule beyond that source is introduced — gaps are explicitly flagged in §25.

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0. How This Document Was Built

This document restructures the functional content of the "Vision AI Enterprise Invoice & Billing Platform — Invoice System — Functional & Test Documentation (v2.7)" into an API-testing-oriented format, following the section layout, per-API specification style, and QA conventions of the "RecruitAI — API Testing Document (v2.0)" used as the structural reference for this exercise.

> **Source-of-Truth Statement:**
>
> The RecruitAI document was used only for structure, section ordering, table layout, and QA terminology. Every functional fact in this document — modules, business rules, fields, endpoints, validation rules, and test scenarios — is sourced exclusively from the Invoice System's own Functional & Test Documentation (v2.7). No endpoint, field, limit, or business rule has been invented. Where the source documentation does not explicitly state an API detail (exact path, verb, or payload shape), this document says so plainly and defers to §25, "Information Missing / To Be Confirmed with Development Team."

Two important scope differences from a typical CRUD API testing document, both directly stated in the source BRD, are carried through consistently:

- The Invoice System does not implement multi-currency conversion or exchange rates. Each invoice is issued in the fixed native currency of its tax regime (₹ for India, ¥ for Japan) — the source BRD (§1.3, Out of Scope) explicitly excludes real-time FX conversion. This document tests currency/tax-regime behaviour accordingly and does not fabricate an exchange-rate API.
- The Invoice System has no discount module, field, or business rule anywhere in the source documentation. No discount section is included in this document; discount-shaped test cases from a generic template would be invented content and are deliberately omitted.

1. Document Information

Project Name	Vision AI Enterprise Invoice & Billing System — Invoice System module
Document Name	Invoice System — API Testing Document
Version	1.0 (derived from Invoice System Functional & Test Documentation v2.7, July 2026)
Prepared By	QA, from the Invoice System BRD/FRD (source of truth)
Reviewed By	To be confirmed with development team
Date	10 August 2026
Purpose	Equip QA engineers, developers, testers, and project leads with a single reference for API understanding, functional validation, negative testing, integration testing, data validation, and regression testing of the Invoice System backend.
Scope	API-level behaviour for all 10 business modules: Authentication, Sender Company ("From") Management, Recipient ("To") Management, Invoice Numbering & Dates, Global Taxation & Bank Autofill, Dynamic Services & Calculation Engine, Form Validation & Cloud Persistence, Invoices Dashboard & Lifecycle, Bank Accounts Management, and Advanced Localization (bilingual PDF / Excel).

2. System Overview

The Invoice System is a bilingual (English / Japanese) invoicing web application within the Vision AI Enterprise Invoice & Billing Platform. It lets an authenticated user configure a sender ("From") company, a recipient ("To") company or employee, an auto-numbered invoice with country-specific taxation and bank details, and a dynamic services/line-item table with a live calculation engine — then persist, edit, preview, download, and email the resulting invoice as a PDF.

2.1 Main Purpose

Remove the numbering drift, tax-compliance risk, and inconsistent bank-detail entry inherent in manual, spreadsheet-based invoicing, by enforcing: atomic per-sender invoice numbering; hard-capped India (CGST/SGST) and Japan (Consumption Tax) tax rates; per-country mandatory bank fields; centralized Cloud MongoDB persistence with a full lifecycle (edit / preview / download / mail / soft-delete); and a single controlled outbound mailbox (hr@visionai.jp) for every dispatched invoice.

2.2 Bilingual Functionality

PDF invoices render in three language modes — English, Japanese, or Both — selectable independently on Download and on Mail dispatch. Japanese rendering applies correct Katakana/Kanji/Hiragana formatting, business-name conventions (e.g. 合同会社 prefix/suffix), and enforced spacing between Japanese and English text. Per BR-10a (confirmed against Commit 0747c74), selecting "Both" always produces two independent single-language PDF files — never one combined bilingual document — on both the Download and Mail paths.

2.3 Multi-Currency / Tax-Regime Functionality

> **Scope Clarification:**
>
> The source BRD (§1.3, Out of Scope) explicitly excludes real-time multi-currency conversion / exchange rates. "Multi-currency" in this system means a per-tax-country fixed currency, not FX conversion: India regime uses ₹ (Indian Rupee) with CGST + SGST (9% each, hard-capped, aggregate 18%); Japan regime uses ¥ (Japanese Yen) with Consumption Tax (10%, hard-capped). Every visible price/total field switches currency symbol and tax fields based on the Tax Country toggle (FR-501–FR-506).

2.4 Main Users / Roles

The source documentation does not confirm a role/permission structure (Admin, Finance/Accounts, HR, Recipient). An earlier draft's assumed-roles table was removed pending dev/product confirmation — see the source document's §2 note. This document therefore treats every authenticated user as having full invoicing rights, matching the documented behaviour, and flags role-based access control as an item to confirm with development (§25).

2.5 High-Level Workflow

- Sign up / log in → JWT issued, user lands on the Dashboard.
- Select Tax Country (India / Japan) → currency symbols and tax fields update.
- Select From (Sender Company): three hardcoded entities (Vision AI, Ideal Folks, VCAS Consulting) or +Others... custom sender.
- Select To (Client): Company or Employee radio, then a saved entry or +Others... manual entry.
- Invoice Number auto-generates from the sender's 3-letter prefix (or a hardcoded prefix); Date defaults to today, Due Date to today + 45 days.
- Bank Details autofill from the sender's (Company→Company) or recipient employee's (Company→Employee) bank profile, optionally overridden via Quick Select.
- Services table: complete row 1 to unlock + Add Item; Subtotal, Tax, Round Off, and Grand Total recompute live.
- Create New Invoice → mandatory-field validation → POST /api/invoices → persisted in Cloud MongoDB → redirect to the Invoices page.
- From the Invoices page: Edit, Preview, Download (English/Japanese/Both), Mail (multi-recipient, fixed template, PDF attachment), or (soft) Delete any invoice.
- Bank Accounts page: maintain reusable bank profiles under India / Japan / International toggles, feeding the Dashboard's Quick Select dropdown.

3. API Testing Scope

This document scopes the following testing activities against the Invoice System backend API, to the extent each is supported by source documentation:

- Functional API testing of every endpoint explicitly documented in the source BRD/FRD (Authentication, Invoices CRUD, Bank Accounts).
- Positive (happy-path) testing against the acceptance criteria of each FR.
- Negative testing — missing mandatory fields, invalid formats, rule violations — derived from BR-01 through BR-12.
- Validation testing of field-level rules from the Data Dictionary (§17).
- Authentication testing — JWT issuance, Bearer-token enforcement, expiry (NFR-S01, NFR-S06).
- Authorization testing — server-side enforcement of sender-email identity (NFR-S02); role-based access control is NOT in scope because it is not confirmed in the source documentation (§2.4).
- CRUD testing for Invoices and Bank Accounts, to the extent each CRUD operation is documented.
- Business rule validation against BR-01 through BR-12.
- Calculation validation — Amount, Subtotal, Tax, Round Off, Grand Total (FR-601–FR-604, BR-08).
- PDF-related API validation — language mode, filename convention, Katakana/Kanji/Hiragana rendering (FR-807, FR-1001, FR-1002, BR-10, BR-10a).
- Bilingual / Japanese character validation — UTF-8 round-trip from form entry through MongoDB to PDF (NFR-L01, NFR-L02).
- Tax-regime (India ₹ / Japan ¥) validation; NOT multi-currency FX conversion, which is out of scope per the source BRD.
- Error handling — to the extent status codes and error shapes are documented; many exact error response bodies are not specified and are flagged in §25.
- Regression testing tied to the numbering, tax, and mandatory-field edge-case checklists in §12/§13 (source §7.1.1–§7.1.3).

Explicitly Out of Scope:

Out of scope for this document, matching the source BRD's own §1.3: payment-gateway integration, automated reminder/dunning emails, real-time FX conversion, and direct integration with external accounting systems (Tally, QuickBooks, Freee, MoneyForward). Load/stress/performance benchmarking and formal penetration testing are not covered here; NFR performance targets are documented for reference in §21.

4. Modules

The ten business modules of the Invoice System, exactly as defined in the source BRD §0 and §1.4. No module has been added, removed, or renamed.

Module ID	Module Name	Description	Key Functions
M-01	Authentication & User Management	Sign-up, login, and JWT session handling for all subsequent API calls.	Register (FR-101); Login/JWT issuance (FR-102); Bearer-token enforcement on /api/* (FR-103, NFR-S01)
M-02	Sender Company (“From”) Management	Maintains the three hardcoded group sender entities and dynamically added custom senders.	List/select sender (FR-201); autofill Logo/Signature/Address/Email (FR-202); persist custom sender on first save (FR-203, BR-11)
M-03	Recipient (“To”) Management	Maintains client Companies and Employees as invoice recipients.	Company vs Employee radio (FR-301); Company dropdown + autofill +

Module ID	Module Name	Description	Key Functions
			persistence (FR-302); Employee manual entry + persistence (FR-303, BR-11)
M-04	Automated Invoice Numbering & Dates	Generates sequential, per-sender-prefixed invoice numbers and date defaults.	3-letter prefix generation (FR-401, BR-03); hardcoded prefixes for the 3 group entities (FR-402); PO Number optional (FR-403); Date/Due Date defaults (FR-404)
M-05	Global Taxation & Bank Autofill	Enforces India (CGST/SGST) or Japan (Consumption Tax) tax regime and the matching mandatory bank-field set.	Tax notification + currency symbol (FR-501, FR-503); rate hard-caps (FR-502, FR-504, BR-01, BR-02); mandatory bank fields per regime (FR-505, FR-506, BR-06); bank autofill source (FR-507); Quick Select (FR-508, BR-12)
M-06	Dynamic Services & Calculation Engine	Line-item table with a live math engine for Amount, Subtotal, Tax, Round Off, and Grand Total.	Row-1 gating (FR-601, BR-07); Amount formula (FR-602); Subtotal/Tax/Round Off/Grand Total formula (FR-603, BR-08); + Add Item / row deletion (FR-604)
M-07	Form Validation & Cloud Persistence	Blocks incomplete submissions and persists valid invoices to Cloud MongoDB.	Mandatory-field validation gate (FR-701, BR-04); POST /api/invoices persistence (FR-702); new invoice visible at top of list (FR-703)
M-08	Invoices Dashboard & Lifecycle	Lists, searches, filters, and drives Edit / Preview / Download / Mail / Delete actions on persisted invoices.	Total counter (FR-801); search (FR-802); date-range filter (FR-803); row action icons (FR-804); Edit→PUT (FR-805); Preview (FR-806); Download (FR-807, BR-10, BR-10a); Mail (FR-809, FR-810, BR-05, BR-10); soft-delete (BR-09, NFR-S03)
M-09	Bank Accounts Management	Maintains reusable bank profiles under India / Japan / International toggles, feeding the Dashboard Quick Select.	3-way toggle + modal (FR-901); per-region mandatory fields (FR-902, FR-903, FR-904, BR-06); Save Account persistence + Quick Select sync (FR-905, BR-12)
M-10	Advanced Localization (Bilingual PDF / Excel)	Japanese-language PDF rendering rules and (per source documentation) an Excel import/export engine still pending full screen-reference documentation.	Katakana conversion, business-name formatting, JA/EN spacing (FR-1001); two-file "Both" output (FR-1002, BR-10a); Excel import/export (excelService.ts) — endpoints not documented, see §25

5. API Architecture / Base URL

The source documentation confirms the backend technology stack and a small number of endpoint paths, but does not publish an explicit base URL, environment list, or API version segment.

Item	Value
Backend framework	Spring Boot 3.x (Java 17+) — confirmed, source §1.5
Persistence	Cloud MongoDB — confirmed, source §1.5
Frontend	React 18 + TypeScript, HTTPS/REST, i18next locale (en/ja) — confirmed, source §1.5
Base URL (Dev)	Not specified in the available documentation – to be confirmed with the development team.
Base URL (Prod)	Not specified in the available documentation – to be confirmed with the development team.
API version segment	Not specified. All confirmed paths (/api/invoices, /api/bank-accounts, /api/auth/login, /api/auth/signup) sit directly under /api with no version prefix — to be confirmed with the development team whether a v1-style prefix exists elsewhere.
HTTP protocol	HTTPS only for credential submission — confirmed (§7.1.4 item 5: “Password fields are transmitted over HTTPS only; no endpoint accepts credentials over plain HTTP.”).
Authentication mechanism	JWT (Bearer token), issued by Spring Security on login and attached to every subsequent /api/* request — confirmed (FR-102, FR-103, NFR-S01).
Content-Type	Not explicitly stated for the JSON endpoints; application/json is the reasonable default given the JSON payload examples in the source (“POST the complete invoice JSON to /api/invoices”, FR-702) — to be confirmed with the development team. File-upload endpoints (Company Logo, Authorised Signature, Excel import) would need multipart/form-data — confirmed as a concept (NFR-S05 references “logo / signature / Excel-import upload endpoints”) but the exact routes are not documented — see §25.
Accept headers	Not specified in the available documentation – to be confirmed with the development team.

6. Endpoint Inventory

Every endpoint below is grouped by module. “Confirmed” entries have an explicit path and verb stated in the source documentation. Entries marked “not documented” in the Endpoint column are business functions that clearly exist in the running system (their UI controls, backend service classes, or data collections are named in the source) but whose exact REST path, verb, and payload shape are not given — these are carried through this document as named gaps rather than invented paths, and are consolidated in §25.

API ID	Module	Method	Endpoint	Purpose	Authentication	Expected Status
API-001	M-01 Authentication	POST	/api/auth/signup	Register a new user	None	200/201 (exact code not documented)
API-002	M-01 Authentication	POST	/api/auth/login	Authenticate and issue a JWT	None (issues the token)	200 (success) / documented failure not specified — see §25
API-003	M-07/M-08 Invoices	POST	/api/invoices	Create (persist) a new invoice	Required (Bearer JWT, per NFR-S01)	200/201 (exact code not documented)
API-004	M-08 Invoices Dashboard	GET	/api/invoices	List invoices for the dashboard (count, search, date filter)	Required (Bearer JWT, per NFR-S01)	200
API-005	M-08 Invoices Dashboard	GET	/api/invoices/{id}	Retrieve a single invoice (inferred — not an explicit source statement; see note below)	Required (Bearer JWT, per NFR-S01)	200 / 404 (not documented)
API-006	M-08 Invoices Dashboard	PUT	/api/invoices/{id}	Update (Save Changes) an existing invoice	Required (Bearer JWT, per NFR-S01)	200
API-007	M-08 Invoices Dashboard	DELETE	/api/invoices/{id}	Soft-delete an invoice (record remains recoverable)	Required (Bearer JWT, per NFR-S01)	200/204 (exact code not documented)
API-008	M-09 Bank Accounts	GET	/api/bank-accounts	List saved bank accounts (feeds the Quick Select dropdown)	Required (Bearer JWT, per NFR-S01)	200
API-009	M-09 Bank Accounts	POST	/api/bank-accounts	Add (Save Account) a new bank account	Required (Bearer JWT, per NFR-S01)	200/201 (exact code not documented)
API-010	M-09 Bank Accounts	PUT	/api/bank-accounts/{id}	Edit a saved bank account (Edit control documented; exact path inferred)	Required (Bearer JWT, per NFR-S01)	200 (not documented)
API-011	M-09 Bank Accounts	DELETE	/api/bank-accounts/{id}	Delete a saved bank account (Delete control documented; exact path inferred)	Required (Bearer JWT, per NFR-S01)	200/204 (not documented)

API ID	Module	Method	Endpoint	Purpose	Authentication	Expected Status
API-012	M-08 Invoices Dashboard	POST (assumed)	Not documented — mail dispatch endpoint	Send Invoice Email (Mail action) — implemented by EmailController.java / apiService.ts per source §1.5 and §3.3.22, but no REST path is given	Required (Bearer JWT, per NFR-S01)	Not documented
API-013	M-08 Invoices Dashboard	GET (assumed)	Not documented — PDF download endpoint	Download invoice PDF (English/Japanese/Both)	Required (Bearer JWT, per NFR-S01)	Not documented
API-014	M-08 Invoices Dashboard	GET (assumed)	Not documented — PDF preview endpoint	Render the Eye-icon Preview modal	Required (Bearer JWT, per NFR-S01)	Not documented
API-015	M-02 Sender Company	Not documented	Not documented — sender CRUD endpoint	Persist a custom (+Others...) sender to the sender collection (BR-11)	Required (Bearer JWT, per NFR-S01)	Not documented
API-016	M-03 Recipient Management	Not documented	Not documented — companies CRUD endpoint	Persist a custom Company recipient to the companies collection (FR-302, BR-11)	Required (Bearer JWT, per NFR-S01)	Not documented
API-017	M-03 Recipient Management	Not documented	Not documented — employees CRUD endpoint	Persist a custom Employee recipient to the dynamicClientEmployees collection (FR-303, BR-11)	Required (Bearer JWT, per NFR-S01)	Not documented
API-018	M-10 Advanced Localization	Not documented	Not documented — Excel import/export endpoint	Import raw billing data / export invoice datasets via excelService.ts	Required (Bearer JWT, per NFR-S01)	Not documented

Inference Flag:

API-005 (GET /api/invoices/{id}) is inferred from the Edit/Preview workflow (FR-805, FR-806) but the source documentation never states this call explicitly — it documents the PUT used on Save Changes (FR-805) but not a corresponding GET. Treat API-005's existence and shape as unconfirmed until verified against the running backend or

API ID	Module	Method	Endpoint	Purpose	Authentication	Expected Status

Postman collection.

7. Detailed Endpoint Documentation

Each confirmed endpoint (API-001 through API-011) is documented below using a consistent field set. Sample request/response payloads use only field names that appear in the source Data Dictionary (§17) — no field has been invented. Where the source does not state a value (exact status code, error shape, header set), the sheet says so explicitly rather than guessing.

7.1 Authentication APIs

API-001 — User Registration (Sign Up)

Module	M-01 Authentication & User Management
Method	POST
Endpoint	/api/auth/signup
Purpose	Creates a new application user account with Full Name, Email, Password, and Confirm Password; validates the two passwords match before creating the account (FR-101).
Authentication	Not required (this is the account-creation endpoint, analogous to the token-issuing Login API).
Headers	Content-Type: application/json (inferred from the JSON payload convention documented for other endpoints — not explicitly stated for this endpoint).
Path Parameters	None.
Query Parameters	None.
Request Body	{ "fullName": string, "email": string, "password": string, "confirmPassword": string }
Request Field Details	fullName — String, free text, required (FR-101, Data Dictionary §17.1). email — String, RFC 5322 format, required. password / confirmPassword — String, minimum 6 characters, required, must match exactly — mismatched values are rejected with a field-level error (FR-101).
Expected Response	Not documented. The source only confirms the resulting UI behaviour: on success the user is redirected directly to the Login page (FR-101 Acceptance Criteria).
Response Field Details	Not documented — to be confirmed with development team.
Expected HTTP Status Codes	Success code not documented (200 or 201 typical — to be confirmed). Password-mismatch and other validation failures are surfaced as a “field-level error” per FR-101; the exact status code is not documented (400 is the reasonable default — to be confirmed).
Business Rules	Password and Confirm Password must match exactly before account creation (FR-101). Password is never stored or transmitted in plaintext beyond the request (Data Dictionary §17.1). Per §7.1.4 item 6 of the source checklist, password should be hashed (not reversibly encrypted) at rest — the exact algorithm (e.g. BCrypt) is not confirmed in the documentation and should be verified with the backend team.
Database Validation	Confirm a new user document is created only when password === confirmPassword. Confirm the stored password value is a hash, never the plaintext value submitted. Collection name is not documented — to be confirmed with development team.

API-002 — Login

Module	M-01 Authentication & User Management
Method	POST
Endpoint	/api/auth/login
Purpose	Authenticates a user by registered Email and Password and issues a JWT token, used as the Bearer credential on every subsequent /api/* request (FR-102, FR-103).

Authentication	Not required (this is the token-issuing endpoint).
Headers	Content-Type: application/json (inferred; not explicitly documented for this endpoint).
Path Parameters	None.
Query Parameters	None.
Request Body	{ "email": string, "password": string }
Request Field Details	email — String, RFC 5322 format, required, must match a registered account. password — String, required.
Expected Response	Not documented in field-level detail. The source confirms only that a JWT token is issued on success (FR-102) and that the token is subsequently attached as a Bearer credential (FR-103).
Response Field Details	token — JWT string, confirmed. Any additional fields (user profile, role, expiry) are not documented — to be confirmed with development team.
Expected HTTP Status Codes	200 on success (JWT issued, user redirected to Dashboard — FR-102 Acceptance Criteria). Invalid credentials: "error shown; no token issued" per FR-102 — exact status code not documented (401 is the reasonable default — to be confirmed with development team).
Business Rules	Only a registered account may authenticate (FR-102). The issued JWT has a defined TTL; expired tokens are rejected server-side and the client is redirected to Login (NFR-S06, TC-EXC-04, → HTTP 401 on an expired token). The JWT payload must not contain the plaintext password or any bank/SWIFT/IFSC data (§7.1.4 item 3).
Database Validation	Confirm the account record queried matches on email; confirm no plaintext password is ever returned in the response body. Collection name not documented — to be confirmed with development team.

Global JWT Enforcement (applies to every API below)

Applies to	Every /api/* endpoint (NFR-S01)
Rule	All /api/* endpoints require a valid JWT Bearer token issued by Spring Security; requests without one return HTTP 401 (NFR-S01, §7.1.4 item 1: "Every /api/* endpoint (invoices, companies, employees, bank-accounts, mail dispatch) rejects requests with a missing or malformed JWT with HTTP 401.")
Header	Authorization: Bearer <token>
Expiry	JWT tokens have a defined TTL (value not documented); an expired token returns HTTP 401 and the UI redirects to Login (NFR-S06, TC-EXC-04).

7.2 Invoice APIs — /api/invoices

API-003 — Create Invoice

Module	M-07 Form Validation & Cloud Persistence
Method	POST
Endpoint	/api/invoices
Purpose	Persists a fully validated invoice to Cloud MongoDB and redirects the user to the Invoices page (FR-702).
Authentication	Required — JWT Bearer token (NFR-S01).
Headers	Authorization: Bearer <jwt>; Content-Type: application/json (inferred).
Path Parameters	None.
Query Parameters	None.
Request Body	{ "senderCompany": "Ideal Folks", "fromEmail": "hr@visionai.jp", "clientType": "COMPANY", "recipientCompany": "Accenture", "recipientAddress": "3-2-1 Marunouchi, Chiyoda-ku, Tokyo", "recipientEmail": "acc@example.com", "companyPhone": "+81-3-1234-5678", "poNumber": "", "date": "2026-08-10", "dueDate": "2026-09-24", "taxCountry": "JAPAN", "consumptionTaxRate": 10, "bankDetails": { "bankName": "GMO Aozora Net Bank", "accountNumber": "1234567", "accountHolderName": "Ideal Folks LLC", "branchName": "Shinjuku Branch", "bankCode": "0397", "branchCode": "001", "swiftCode": "", "accountType": "SAVINGS" }, "services": [{ "serviceDescription": "Consulting — August", "hours": 23, "unitPrice": 345 }] }
Request Field Details	senderCompany — Enum/String, required; Vision AI | Ideal Folks | VCAS Consulting | custom (Data Dictionary §17.1). fromEmail — String, RFC 5322, required; cosmetic display only — actual SMTP dispatch always uses hr@visionai.jp regardless of this value (BR-05, NFR-S02). clientType — Enum, required; COMPANY | EMPLOYEE, defaults to COMPANY (FR-301). recipientCompany / recipientEmployee — String/Object, required; drives invoice-number prefix (FR-401) and bank autofill source (FR-507). recipientAddress — String, required. recipientEmail — String, RFC 5322, required. companyPhone / employeePhone — String, required when the recipient was entered via +Others.... invoiceNumber — NOT part of the request; system-generated, read-only, must never accept manual entry (BR-03). poNumber — String, optional (poNumber?), may be empty (FR-403). date / dueDate — Date (YYYY-MM-DD), required; date defaults to today, dueDate defaults to date + 45 days, both editable (FR-404). taxCountry — Enum, required; INDIA | JAPAN (§17.3). cgstRate / sgstRate — Decimal 0–9, required under India; default 9 each, values above 9 must be rejected or clamped server-side (FR-502, BR-01, NFR-S04). consumptionTaxRate — Decimal 0–10, required under Japan; default 10, values above 10 must be rejected or clamped server-side (FR-504, BR-02, NFR-S04). bankDetails.* — required fields vary by taxCountry per BR-06 (see §9 Invoice-Specific API Validations). services[] — array, at least one row required; each row needs serviceDescription (String, required for row 1), hours (Decimal ≥ 0, must be > 0 for row 1 to satisfy the Add-Item gate, FR-601/BR-07), unitPrice (Decimal ≥ 0, same > 0 rule).
Expected Response	Not documented in field-level detail. The source confirms only the resulting UI behaviour: response is treated as success/failure ("Response 201/200" is the assumption carried in the source's own TC-PERSIST-01 test case, not a documented contract) and the created invoice appears at the top of the Invoices list with Invoice Number, Client (Name & Email), Date Created, and Amount visible immediately (FR-703).
Response Field Details	invoiceNumber — system-generated, format INV-[PREFIX]-[n] (FR-401/FR-402/BR-03). dateCreated — ISO 8601 timestamp, set on first successful POST (§17.2). Remaining response shape not documented — to be confirmed with development team.
Expected HTTP Status Codes	200 or 201 on success (exact code not documented — the source's own test case TC-PERSIST-01 notes "POST /api/invoices succeeds" without pinning a code). 400-class

	validation failure implied by BR-04/FR-701 ("submission blocked ... no POST is issued" client-side; server-side rejection code not documented).
Business Rules	BR-01 (CGST/SGST cap 9% each), BR-02 (Consumption Tax cap 10%), BR-03 (auto-numbering, read-only), BR-04 (mandatory-field gate — no partial invoice may be persisted), BR-06 (per-regime mandatory bank fields), BR-07 (Services row-1 gating), BR-08 (Grand Total = Subtotal + Tax + Round Off, math engine authoritative), BR-11 (custom sender/company/employee persisted to their collection on first save).
Database Validation	Confirm the invoice document is written only when every BR-04 mandatory field is present. Confirm cgstRate/sgstRate/consumptionTaxRate are never stored above their caps even if a raw request attempts to bypass the client-side clamp (NFR-S04). Confirm grandTotal stored equals subtotal + taxAmount(s) + roundOff exactly. Confirm a new custom sender/company/employee is upserted into its respective collection (sender / companies / dynamicClientEmployees) per BR-11. Collection name for invoices themselves is not documented — to be confirmed with development team.

API-004 — List Invoices

Module	M-08 Invoices Dashboard & Lifecycle
Method	GET
Endpoint	/api/invoices
Purpose	Returns the invoices shown on the Invoices Dashboard, backing the Total Invoices counter, the search box, and the date-range filter (FR-801, FR-802, FR-803).
Authentication	Required — JWT Bearer token (NFR-S01).
Headers	Authorization: Bearer <jwt>; Accept: application/json (inferred).
Path Parameters	None.
Query Parameters	Not documented with exact parameter names. The source confirms the dashboard supports a free-text search (Invoice Number, Employee/Company Name, or Email — FR-802) and a From/To created-date range filter (FR-803), but does not name the query-string parameters used to implement them — to be confirmed with development team. Pagination parameters, if any, are not documented (NFR-P02 references "server-side pagination/filtering" for lists up to 1,000 invoices, implying pagination exists, but no parameter names are given).
Request Body	None.
Response Field Details	Each list row must expose at minimum: Invoice Number, Client Name & Email, Date Created, and Amount (FR-703). Full response envelope shape (e.g. whether it is a bare array or a paged object) is not documented — to be confirmed with development team.
Expected HTTP Status Codes	200 on success. Additional codes not documented.
Business Rules	The Total Invoices counter must reflect the current live (non-deleted) count in MongoDB (FR-801). Soft-deleted invoices must be excluded from this list (BR-09).
Database Validation	Confirm totalInvoicesCount (or equivalent) matches the count of ACTIVE (non-SOFT_DELETED) invoice documents (§17.2, §17.3).

API-005 — Get Invoice by ID (inferred)

Module	M-08 Invoices Dashboard & Lifecycle
Method	GET
Endpoint	/api/invoices/{id}
Purpose	Presumed retrieval of a single invoice's full detail, to support the Edit (Pencil) and Preview (Eye) actions. NOT an explicit statement in the source documentation — the source

	documents only the PUT used by Save Changes (FR-805). Treat this endpoint's existence and contract as unconfirmed.
Authentication	Required — JWT Bearer token (NFR-S01), if the endpoint exists as inferred.
Path Parameters	id — invoice identifier.
Expected HTTP Status Codes	Not documented.
Business Rules	Not documented for this specific call.
Database Validation	Not documented.
QA Note	Before writing automated tests against this endpoint, confirm its existence, path, and response shape against the running backend or the team's Postman collection. Do not assume the shape below is correct — API details to be confirmed with development team.

API-006 — Update Invoice (Save Changes)

Module	M-08 Invoices Dashboard & Lifecycle
Method	PUT
Endpoint	/api/invoices/{id}
Purpose	Updates an existing invoice when the user edits it (Pencil icon) and clicks Save Changes; updates the MongoDB record and the PDF artifacts, then returns the user to the Invoices page with the updated row visible (FR-805).
Authentication	Required — JWT Bearer token (NFR-S01).
Headers	Authorization: Bearer <jwt>; Content-Type: application/json (inferred).
Path Parameters	id — invoice identifier.
Request Body	Same shape as Create Invoice (API-003) with the edited fields — not separately documented beyond “PUT to /api/invoices/{id}” (FR-805).
Request Field Details	Same field set and validation rules as API-003 apply on edit: “Editing a saved invoice (Pencil icon) re-validates all mandatory fields exactly as Create does; Save Changes cannot bypass validation that Create enforces” (source §7.1.3 item 5, mapped to FR-805/BR-04).
Expected Response	Not documented in field-level detail.
Expected HTTP Status Codes	200 on success (not explicitly pinned in the source; inferred from standard PUT semantics).
Business Rules	BR-04 (mandatory-field re-validation applies identically to edit). Editing must update both the MongoDB record and the associated PDF artifacts (FR-805).
Database Validation	Confirm the updated fields persist and that the downloaded/emailed PDF reflects the change (“Editing a field and saving reflects the change on the row and on the downloaded PDF”, FR-805 Acceptance Criteria).

API-007 — Delete Invoice (Soft Delete)

Module	M-08 Invoices Dashboard & Lifecycle
Method	DELETE
Endpoint	/api/invoices/{id}
Purpose	Removes an invoice from the active Invoices list via the Trash icon and confirmation prompt. Confirmed as a SOFT delete only — the record must remain recoverable (BR-09, NFR-S03).
Authentication	Required — JWT Bearer token (NFR-S01).
Headers	Authorization: Bearer <jwt>.

Path Parameters	id — invoice identifier.
Expected Response	Not documented in field-level detail.
Expected HTTP Status Codes	Not documented (200/204 typical for a delete — to be confirmed with development team).
Business Rules	BR-09: once persisted, an invoice may only be soft-deleted from the Invoices page; direct database hard-deletes are not exposed via any API endpoint, and the record must remain recoverable in audit history. NFR-S03: "DELETE /api/invoices/{id} either performs a soft-delete or is rejected; record remains recoverable." Test case TC-DASH-N01 in the source explicitly targets a raw DELETE request bypassing the confirmation modal, expecting the record to be soft-deleted only (or the request rejected) and never physically purged.
Database Validation	Confirm the invoice document still exists in MongoDB after DELETE, with its status changed to SOFT_DELETED (§17.3) rather than the document being removed. Confirm the record is excluded from GET /api/invoices (API-004) and from the totalInvoicesCount (FR-801) while remaining queryable for audit purposes.

7.3 Bank Accounts APIs — /api/bank-accounts

API-008 — List Bank Accounts

Module	M-09 Bank Accounts Management
Method	GET
Endpoint	/api/bank-accounts
Purpose	Returns saved bank accounts, populating both the Bank Accounts page table and the Dashboard's Quick Select dropdown (FR-508, FR-905, BR-12). Endpoint path is explicitly named in the source's own exception test case TC-EXC-06 (“Simulate /api/bank-accounts timeout”).
Authentication	Required — JWT Bearer token (NFR-S01).
Headers	Authorization: Bearer <jwt>; Accept: application/json (inferred).
Path Parameters	None.
Query Parameters	Not documented — to be confirmed with development team.
Expected HTTP Status Codes	200 on success. TC-EXC-06 confirms the expected degraded behaviour on a network timeout: “Dropdown shows an empty/error state rather than hanging indefinitely; manual bank entry remains available as a fallback” — exact timeout/error status code not documented.
Business Rules	BR-12: newly saved accounts must appear in Quick Select immediately, with no page reload required.
Database Validation	Confirm every document in the bank_accounts collection (§17.9, FR-905) with the requesting user's scope (if any — scoping is not documented) is returned.

API-009 — Add Bank Account (Save Account)

Module	M-09 Bank Accounts Management
Method	POST
Endpoint	/api/bank-accounts
Purpose	Persists a new bank account to the bank_accounts collection, displays it in the Bank Accounts table with Edit/Delete controls, and immediately populates the Dashboard Quick Select dropdown (FR-905).
Authentication	Required — JWT Bearer token (NFR-S01).
Headers	Authorization: Bearer <jwt>; Content-Type: application/json (inferred).
Request Body	{ "region": "JAPAN", "bankName": "GMO Aozora Net Bank", "accountNumber": "1234567", "accountHolderName": "Ideal Folks LLC", "branchName": "Shinjuku Branch", "bankCode": "0397", "branchCode": "001", "swiftCode": "", "accountType": "SAVINGS" }
Request Field Details	region — Enum, required; INDIA | JAPAN | INTERNATIONAL (3-way toggle, FR-901; bankRegime enum, §17.3). bankName, accountNumber, accountHolderName, branchName, accountType — String/Enum, required in all three regions (BR-06, Data Dictionary §17.1). ifscCode — String, required under India only; not shown/required under Japan or International-only flows (BR-06). bankCode, branchCode — String, required under Japan and International; Japan Zengin System codes (FR-506, FR-904). swiftCode — String, 8 or 11-character SWIFT/BIC format; NOT mandatory under India or Japan, becomes mandatory only under International (FR-904, BR-06).
Expected HTTP Status Codes	Not documented (200/201 typical — to be confirmed).

Business Rules	FR-901–FR-904 (per-region mandatory-field enforcement, mirroring BR-06 exactly as it applies on the Dashboard). FR-905 / BR-12 (immediate Quick Select sync, no reload).
Database Validation	Confirm a document is written to bank_accounts only when every region-mandatory field is present. Confirm the account is retrievable via API-008 (GET /api/bank-accounts) immediately after save, with no caching delay.

API-010 / API-011 — Edit / Delete Bank Account

Module	M-09 Bank Accounts Management
Method	PUT (Edit) / DELETE
Endpoint	/api/bank-accounts/{id} (inferred — the source documents Edit and Delete as table row controls, FR-905, but does not state the REST path or verb explicitly).
Purpose	Edit or remove a saved bank account from the Bank Accounts table, keeping the Quick Select dropdown in sync.
Authentication	Required — JWT Bearer token (NFR-S01), if these endpoints exist as inferred.
Path Parameters	id — bank account identifier.
Field Validations	Same per-region mandatory-field rules as API-009 apply on Edit.
Expected HTTP Status Codes	Not documented.
Business Rules	BR-12: any change must be reflected in Quick Select without a page reload.
Database Validation	Not documented.
QA Note	The existence of a dedicated Edit/Delete REST path is a reasonable inference from the documented UI controls, but is not itself confirmed. API details to be confirmed with development team before automating.

7.4 Endpoints Referenced by the BRD but Not Documented at the API Level

The following business functions are clearly real — each is named via its UI control, backend service class, or MongoDB collection in the source documentation — but the source never states their REST path, HTTP verb, or payload shape. Per the instructions for this document, no path is invented; each is recorded here as a confirmed business function with an unconfirmed API contract.

API-012 — Mail Dispatch (Send Invoice Email)

Module	M-08 Invoices Dashboard & Lifecycle
Endpoint	Not documented — API details to be confirmed with development team.
Known implementation	apiService.ts (frontend) and EmailController.java (backend), per source §1.5 and §3.3.22/§7.3.
Purpose (confirmed)	Dispatches the invoice email from a single sender (hr@visionai.jp) to a fixed Primary Recipient (the invoice's own Bill To email, read-only in the Mail modal) plus optional comma-separated Additional Recipients, with a PDF attachment in the selected language (FR-809, FR-810, BR-05, BR-10, BR-10a).
Request Field Details (confirmed, shape not documented)	primaryRecipientEmail — system-derived, read-only, fixed to the invoice's Bill To email; not editable from the Mail modal (Data Dictionary §17.1). additionalRecipients — String, optional, comma-separated email list, parsed on commas (Data Dictionary §17.1, §7.1.5 item 3). mailLanguage — Enum, required; English | Japanese | Both (FR-809). “Both” always attaches two separate single-language PDFs, never one bilingual file (BR-10a).
Business Rules	BR-05: every outbound email originates from hr@visionai.jp at the SMTP envelope level regardless of the invoice's displayed sender, enforced server-side by BrevoEmailService.resolveSenderEmail() — confirmed by dev. BR-10 / BR-10a: attachment

	filename convention. FR-810: fixed email body template — header, greeting with client name, month extracted from invoice date, line-item table, Subtotal/Tax/Round Off/Grand Total block, and Payment Instructions block; no custom body may be substituted at the client.
Exception behaviour	A failed SMTP send must not silently mark the invoice as "sent" in any UI state (§7.1.5 item 6, TC-EXC-02); dispatch fails gracefully with a clear error, the invoice record is unaffected, and no duplicate emails are sent on retry.

API-013 / API-014 — PDF Download / Preview

Module	M-08 Invoices Dashboard & Lifecycle
Endpoint	Not documented — API details to be confirmed with development team.
Known implementation	PDF Generator (iText / HTML-to-PDF / PDF-LIB) server-side generation, streamed to client; katakanaConverter.ts for Japanese phonetic conversion (source §1.5).
Purpose (confirmed)	Download: Arrow icon opens a language dropdown (English / Japanese / Both); English/Japanese each download a single PDF, Both triggers two simultaneous downloads (FR-807, BR-10a). Preview: Eye icon opens an interactive modal rendering the PDF in place without triggering a file download (FR-806).
Business Rules	BR-10 / BR-10a filename convention applies identically to Download. FR-1001: Japanese PDFs must render correct Katakana/Kanji/Hiragana with no garbled characters. NFR-A03: a PDF generation failure (e.g. malformed Japanese font asset) must not corrupt the underlying invoice record; the failing language's download fails cleanly while other languages remain unaffected (TC-EXC-03).

API-015 through API-018 — Sender / Company / Employee / Excel Endpoints

Endpoints	Not documented — API details to be confirmed with development team.
Known implementation	Sender persistence: sender collection (BR-11). Company recipients: companies collection (§1.3, BR-11). Employee recipients: dynamicClientEmployees collection (§1.3, BR-11, Glossary §17.6). Excel import/export: excelService.ts (source §1.5); per the source's own Resolution Log (§7.3), Module 10's Excel import/export screens still lack screen-reference documentation as of v2.7, though development has confirmed the underlying engine is "fully tested and functional."
Purpose (confirmed)	Custom senders, companies, and employees added via +Others... are persisted to their respective collection on first invoice save so they appear in their dropdown on subsequent invoices, displayed thereafter as a read-only summary card rather than re-opening the manual entry fields (BR-11, FR-203, FR-302, FR-303). Excel import brings in raw billing data; Excel export produces invoice datasets (§1.5).
QA Note	Do not write automated API tests against a guessed path for these functions. Confirm exact endpoints with the development team, then extend §6/§7 of this document accordingly.

8. Invoice-Specific API Validations

Validation rules pulled directly from the Business Rules (§3.1 of the source), Functional Requirements, and Data Dictionary. The source Invoice System has no discount module, so no discount validations are listed — including one would be invented content.

8.1 Recipient (Company / Employee)

- Required fields: clientType (radio, defaults COMPANY — FR-301); recipientAddress and recipientEmail are both mandatory once a recipient is chosen or entered (FR-302, FR-303).
- Company +Others...: Company Name, Company Email, Company Address (*), Company Phone (*) required on manual entry (FR-302); leaving Company Address empty blocks submission (TC-CLIENT-N01).
- Employee first-time flow: Employee Name, Employee Email, Employee Address (*), Employee Phone required (FR-303).
- Duplicate handling: not documented — the source does not state whether a duplicate Company/Employee name or email is rejected, merged, or allowed. To be confirmed with development team.
- Update / deletion of a saved Company or Employee via the API: not documented. The source only confirms persistence on first save (BR-11); no update or delete flow for recipients is described.
- Invalid recipient data: an empty mandatory field on the +Others... path must block submission and show “This field is mandatory” (BR-04, FR-701).

8.2 Invoice

- Invoice creation: every BR-04 mandatory field must be present or the POST is blocked client-side; the source does not confirm the exact server-side rejection behaviour beyond NFR-S04's tax-cap guarantee.
- Invoice number: system-generated, read-only, sequential per sender prefix; must never accept manual entry (BR-03, FR-401, FR-402). Numbering is scoped per prefix, not global — INV-ACC-*, INV-VI-*, INV-IF-*, and INV-VCAS-* sequences increment independently.
- Invoice date: defaults to today (YYYY-MM-DD), editable via calendar picker (FR-404).
- Due date: defaults to date + 45 days, editable independently — changing Date does not automatically shift an already-set Due Date (FR-404, §3.3.3 Test Focus).
- Customer selection: drives both the invoice-number prefix (FR-401) and the bank-detail autofill source (FR-507): Company→Company autofills from the sender's bank; Company→Employee autofills from the recipient employee's bank.
- Invoice items (services[]): at least one row required; row 1 must have a valid Description, Hours > 0, and Unit Price > 0 before + Add Item is enabled (BR-07, FR-601).
- Quantity / Hours: Decimal ≥ 0; must be > 0 for row 1 to satisfy the Add-Item gate; accepts fractional values (TC-CALC-E03: Hours = 7.5 → Amount 750.00).
- Unit price: Decimal ≥ 0, same > 0 gating rule as Hours for row 1.
- Subtotal: sum of every row Amount (FR-603).
- Discount: NOT APPLICABLE. No discount field, rule, or FR exists anywhere in the source documentation.
- Tax: see §8.3 below.
- Grand total: Subtotal + Tax Amount + Round Off, computed from exact underlying decimals; hand-typed totals are never accepted — the math engine is authoritative (BR-08).
- Currency: see §8.3 below.
- Notes: no “notes” field is documented anywhere in the source Data Dictionary or FR tables. Not applicable.
- Status: implicit invoiceStatus enum ACTIVE | SOFT_DELETED (§17.3) — ACTIVE invoices are listed and counted; SOFT_DELETED invoices are hidden from the list but remain in MongoDB for audit (BR-09).

8.3 Currency / Tax Regime

> **Scope Reminder:**
>
> This system does not implement multi-currency conversion. "Currency" validation here means the fixed India ₹ / Japan ¥ regime tied to the Tax Country toggle — not FX rates or a converted-amount field. There is no exchangeRate field in the Data Dictionary.

- Supported currencies: ₹ (India regime) and ¥ (Japan regime) only — derived automatically from taxCountry, never entered directly (FR-501, FR-503, currencySymbol field §17.2).
- Currency code: taxCountry enum values are INDIA | JAPAN (§17.3); there is no separate ISO currency-code field documented.
- Currency symbol: system-derived (₹ or ¥); every visible price/total field (Unit Price, Amount, Subtotal, tax line(s), Round Off, Grand Total) must switch symbol together when Tax Country is toggled, without losing already-entered Hours/Unit Price values (§7.1.2 item 6).
- Currency conversion / exchange rate: OUT OF SCOPE per the source BRD §1.3. No conversion endpoint, field, or rate exists to validate.
- Rounding: Round Off is the signed decimal adjustment (positive or negative) that makes Grand Total a whole integer in the display currency (§17.2, BR-08); the sign must be preserved correctly through save, edit, PDF render, and email body (§7.1.2 item 3).
- CGST / SGST (India): default 9% each, hard-capped at 9 — values above 9 must clamp down both client-side (on blur) and server-side (NFR-S04); values below 9 are accepted as entered; a boundary value of exactly 9 is accepted with no clamping event (TC-TAX-E01).
- Consumption Tax (Japan): default 10%, hard-capped at 10, same clamp behaviour as above (TC-TAX-E03 for the boundary case).
- Zero-value tax rates (0%) are accepted and must produce a ₹0.00 / ¥0.00 tax line without breaking Grand Total math (TC-TAX-E04).
- Negative tax-rate entry: must be rejected or clamped to 0; the source flags this as a case to verify rather than a confirmed pass (TC-TAX-E05).
- Multiple tax scenarios: India applies CGST and SGST as two independent components with independent 9% caps — raising one does not affect the other's cap (§7.1.2 item 1). India and Japan tax fields are mutually exclusive on the Tax Country toggle (§7.1.2 item 2); the system does not support applying both regimes to one invoice.

8.4 Discount

NOT APPLICABLE. The Invoice System's source documentation contains no discount field, business rule, or functional requirement of any kind. This section is intentionally left without validations rather than inventing discount behaviour that does not exist in the system.

8.5 Japanese / English (Bilingual)

- English characters: baseline; every UI label and PDF template must have both an English and Japanese translation key (NFR-L03).
- Japanese Hiragana / Katakana / Kanji: all three writing systems must render correctly on Japanese invoice PDFs (Glossary §17.6); Katakana conversion is applied via katakanaConverter.ts for service descriptions and company titles where specified (FR-1001).
- Mixed English + Japanese text: addSpaceBetweenJaAndEn enforces professional spacing between Japanese characters and adjacent English alphanumeric text on rendered PDFs (NFR-L02).
- Japanese customer/company names: business-Japanese formatting conventions (e.g. 合同会社 prefix/suffix) must be applied rather than a literal English-to-Japanese field copy (FR-1001, §7.1.6 item 6).
- Japanese addresses: Kanji / Hiragana must be preserved through form entry, MongoDB persistence, and PDF rendering (FR-1001).

- UTF-8 encoding: all text fields (bank name, branch name, addresses, descriptions) must support and correctly round-trip UTF-8 Japanese characters (Kanji, Hiragana, Katakana, full-width Latin) end-to-end — zero mojibake / replacement-character defects across a Japanese-data regression pass (NFR-L01).
- Data persistence: same UTF-8 round-trip requirement applies to the MongoDB layer specifically (NFR-L01).
- API response encoding: not separately documented beyond the general UTF-8 requirement in NFR-L01 — to be confirmed with development team whether API responses declare charset=utf-8 explicitly.
- Database storage: verify stored field values are byte-identical Japanese text, not a transliterated or corrupted form.
- PDF rendering: Bilingual PDFs must present both languages without truncation or overlapping text blocks at standard A4/Letter page sizes (§7.1.6 item 7); "Both" always produces two separate single-language PDFs, never one combined bilingual document (BR-10a, FR-1002).

Important Test:

Japanese text must remain unchanged through FORM → API → DATABASE → API RESPONSE → PDF. Example — Input: 山田太郎 → Database: 山田太郎 → API Response: 山田太郎 → PDF: 山田太郎. Any of ???, □□□, or garbled characters at any stage is a defect.

9. API Test Scenarios

Every scenario below carries forward the source document's own Test Case ID(s) so this API-testing document stays traceable to the original BRD/FRD test matrix (§4 of the source) rather than introducing a parallel numbering scheme.

Scenario ID	Module	Scenario	Type	Source Test Case ID(s)
SC-01	M-01 Authentication	Sign up with matching passwords, then log in and confirm JWT is issued and accepted on a protected call	Positive	TC-AUTH-01, TC-AUTH-02
SC-02	M-01 Authentication	Call any /api/* endpoint with no Authorization header	Security / Negative	TC-AUTH-N02 (§7.1.4 item 1)
SC-03	M-01 Authentication	Call a protected endpoint with an expired JWT	Negative / Exception	TC-EXC-04
SC-04	M-04 Numbering	Create an invoice for a new (non-hardcoded) recipient company and confirm the 3-letter prefix + sequence	Positive	TC-INV-GEN-01
SC-05	M-04 Numbering	Create an invoice for each of the 3 hardcoded senders and confirm the fixed prefix	Positive	TC-INV-GEN-02
SC-06	M-04 Numbering	Recipient company name shorter than 3 letters / starting with a digit or symbol	Edge	TC-INV-GEN-E01, TC-INV-GEN-E02
SC-07	M-04 Numbering	Two differently-named senders whose names collapse to the same 3-letter prefix	Edge	§7.1.1 item 3
SC-08	M-04 Numbering	Concurrent invoice creation for the same prefix	Edge / Load	NFR-S08, §7.1.1 item 4
SC-09	M-05 Taxation	Enter CGST/SGST above the 9% cap under India regime	Negative	TC-TAX-N01
SC-10	M-05 Taxation	Enter Consumption Tax above the 10% cap under Japan regime	Negative	TC-TAX-N02
SC-11	M-05 Taxation	Enter CGST/Consumption Tax exactly at the cap boundary (9 / 10)	Edge (boundary)	TC-TAX-E01, TC-TAX-E03
SC-12	M-05 Taxation	Enter a 0% tax rate	Edge (boundary)	TC-TAX-E04
SC-13	M-05 Taxation	Enter a negative tax rate	Negative	TC-TAX-E05
SC-14	M-05 Bank Validation	Submit an India-regime invoice with IFSC Code empty	Negative	TC-BANK-N01
SC-15	M-05 Bank Validation	Submit a Japan-regime invoice with Bank Code empty	Negative	TC-BANK-N02
SC-16	M-05 Bank Validation	Submit an International bank account (Bank Accounts page) with SWIFT Code empty	Negative	TC-BANKPAGE-N01
SC-17	M-03 Recipient	Add a new Company recipient via +Others... with Company Address left empty	Negative	TC-CLIENT-N01
SC-18	M-06 Calculation	Click + Add Item before row 1 is complete	Negative	TC-SVC-N01
SC-19	M-06 Calculation	Hours = 0 / very large Hours & Unit Price / fractional Hours	Edge (boundary)	TC-CALC-E01, TC-CALC-E02, TC-CALC-E03
SC-20	M-07 Validation	Submit the Create Invoice form with exactly one mandatory field left empty	Negative	TC-VALID-N01
SC-21	M-07 Persistence	Submit a fully valid invoice end-to-end	Positive	TC-PERSIST-01

Scenario ID	Module	Scenario	Type	Source Test Case ID(s)
SC-22	M-08 Dashboard	Send a raw DELETE request to /api/invoices/{id} bypassing the confirmation modal	Security / Negative	TC-DASH-N01
SC-23	M-08 Mail	Send an invoice email with a custom “From Email” other than hr@visionai.jp and inspect the SMTP envelope	Security / Negative	TC-MAIL-N01
SC-24	M-08 Mail	Send an invoice email and inspect the attachment filename against the naming convention	Positive / Negative	TC-MAIL-N02
SC-25	M-08 Mail	Send an invoice email in Production (default tenant) and confirm the sender header	Positive (regression)	TC-MAIL-N03
SC-26	M-08/M-09 Integration	Save a Bank Account, then confirm it appears in Quick Select without reload; select it and confirm it overrides the Dashboard bank fields	Integration	§3.3.18, FR-508, BR-12
SC-27	M-07 Exception	MongoDB unreachable during Create Invoice save	Exception	TC-EXC-01
SC-28	M-08 Exception	SMTP/Brevo unavailable during Mail dispatch	Exception	TC-EXC-02
SC-29	M-08 Exception	PDF generation service throws an error (e.g. malformed Japanese font asset)	Exception	TC-EXC-03
SC-30	M-09 Exception	GET /api/bank-accounts times out while opening Quick Select	Exception	TC-EXC-06
SC-31	M-01/M-08 Regression	Full end-to-end regression: sign up → login → create invoice (each tax regime) → edit → preview → download (EN/JA/Both) → mail → soft-delete	Regression	§12 End-to-End Workflow

10. API Test Cases

Detailed test cases, grouped by area, reusing the source document's own TC-IDs. Every row is traceable back to its FR/BR in §3 of the source and to the Scenario table in §9 above.

10.1 Authentication

TC ID	Module	API	Test Scenario	Preconditions	Test Data	Steps	Expected Result	Priority
TC-AUTH-01	M-01	POST /api/auth/signup	Sign up with matching passwords	User not registered	fullName=Test User, email=test@vai.jp, password=Passw0rd!, confirmPassword=Passw0rd!	1. POST the signup payload	Account created; user redirected to Login page (exact status code not documented)	High
TC-AUTH-02	M-01	POST /api/auth/login	Login with valid credentials	Registered account exists	Valid email + password	1. POST the login payload	JWT issued; user redirected to Dashboard	High
TC-AUTH-N01	M-01	POST /api/auth/login	Login with wrong password	Registered account exists	Valid email, wrong password	1. POST the login payload	Error shown; no token issued (exact status/body not documented — confirm)	High
TC-AUTH-N02	M-01	Any /api/*	Call a protected endpoint with no JWT	None	No Authorization header	1. Call e.g. GET /api/invoices with no token	HTTP 401 (NFR-S01)	High
TC-EXC-04	M-01	Any /api/*	Call a protected endpoint with an expired JWT	Valid session, token TTL elapsed	Expired JWT	1. Wait for expiry or force-expire in staging 2. Call any protected endpoint	HTTP 401; UI redirects to Login with a session-expired message; in-progress unsaved form data is not silently lost without warning	Medium

10.2 CRUD — Invoices, Recipients, Bank Accounts

TC ID	Module	API	Test Scenario	Preconditions	Test Data	Steps	Expected Result	Priority
TC-INV-GEN-01	M-04	POST /api/invoices	Auto-number for a non-hardcoded company	Recipient = new company “Accenture”	Company Name = Accenture	1. Select Accenture as To (Client) 2. Observe Invoice Number field	Invoice Number auto-populates INV-ACC-1 (or next sequence)	High
TC-INV-GEN-02	M-04	POST /api/invoices	Hardcoded prefix for Vision AI sender	Sender = Vision AI	Sender = Vision AI	1. Select Vision AI as From 2. Observe Invoice Number field	Invoice Number auto-populates INV-VI-<n>	High
TC-TAX-N01	M-05	POST /api/invoices	CGST entered above the India cap	Tax Country = India	CGST input = 15	1. Enter CGST = 15 2. Blur/submit	Value clamps to 9 on blur/save; stored value never exceeds 9 (NFR-S04 requires server-side enforcement too)	High
TC-TAX-N02	M-05	POST /api/invoices	Consumption Tax entered above the Japan cap	Tax Country = Japan	Tax input = 20	1. Enter Consumption Tax = 20 2. Blur/submit	Value clamps to 10; stored value never exceeds 10	High
TC-TAX-E01	M-05	POST /api/invoices	CGST entered at exactly the cap boundary	Tax Country = India	CGST = 9	1. Enter CGST = 9 exactly	Value accepted as-is (inclusive boundary); no clamping event	Medium
TC-TAX-E04	M-05	POST /api/invoices	CGST entered as 0	Tax Country = India	CGST = 0	1. Enter CGST = 0	Value accepted; CGST tax line = ₹0.00; SGST unaffected	Low
TC-BANK-N01	M-05	POST /api/invoices	India invoice missing IFSC Code	Tax Country = India; all other bank fields filled	IFSC Code = ""	1. Leave IFSC Code empty 2. Submit	Submission blocked; scroll/focus to IFSC Code; “This field is mandatory” shown	High
TC-BANK-N02	M-05	POST /api/invoices	Japan invoice missing Bank Code	Tax Country = Japan; all other bank fields filled	Bank Code = ""	1. Leave Bank Code empty 2. Submit	Submission blocked; scroll/focus to Bank Code; “This field is mandatory” shown	High
TC-CLIENT-N01	M-03	POST /api/invoices (or company-	New company missing	Company +Others... selected	Company Address = ""	1. Fill Name/Email only 2.	Submission blocked; scroll/focus to	High

TC ID	Module	API	Test Scenario	Preconditions	Test Data	Steps	Expected Result	Priority
		persistence endpoint, not documented)	mandatory Address			Leave Company Address empty 3. Submit	Company Address; error shown	
TC-SVC-N01	M-06	POST /api/invoices	+ Add Item clicked before row 1 complete	Fresh Services table (row 1 empty)	Row 1 = all empty	1. Attempt to click + Add Item while row 1 is incomplete	+ Add Item remains disabled; no second row is added	Medium
TC-VALID-N01	M-07	POST /api/invoices	Submit with any one mandatory field missing	Any one * field left empty	One * field blank (e.g. Employee Phone)	1. Complete all fields except one * field 2. Submit	Submission blocked; auto-scroll/focus to the exact missing field; “This field is mandatory” inline error; no POST issued	High
TC-PERSIST-01	M-07	POST /api/invoices	Full valid invoice payload persists	All mandatory fields complete and valid	Complete valid invoice payload	1. Complete every * field 2. Submit	POST /api/invoices succeeds; new invoice visible at top of Invoices list with correct Invoice Number, Client, Date Created, Amount	High
TC-DASH-N01	M-08	DELETE /api/invoices/{id}	Attempt hard-delete via direct API call	Invoice exists	Direct API DELETE /api/invoices/{id}	1. Send a raw DELETE request bypassing the confirmation modal	Record is soft-deleted only (or the request is rejected without the proper flow); record remains recoverable/auditable, never physically purged	High
TC-MAIL-N01	M-08	Mail dispatch endpoint (not documented)	Attempt to override sender identity	Invoice with a custom “From Email” other than hr@visionai.jp	Invoice fromEmail = custom@othersender.com	1. Trigger Mail send 2. Inspect the outbound SMTP envelope “From” header	Outbound envelope “From” is still hr@visionai.jp regardless of the invoice's displayed sender email (BR-05)	High
TC-MAIL-N02	M-08	Mail dispatch endpoint (not documented)	Attachment filename does not match convention	Invoice Number and Description set	Invoice Number=INV-VI-5, Description=Vijayalakshmi M	1. Send email 2. Inspect attachment filename	Filename is exactly “INV-VI-5 Vijayalakshmi M.pdf” — any deviation is a defect (BR-10)	Medium
TC-BANKPAGE-N01	M-09	POST /api/bank-accounts	International account	International toggle selected	SWIFT Code = ""	1. Fill all fields except	Submission blocked; SWIFT Code flagged as	High

TC ID	Module	API	Test Scenario	Preconditions	Test Data	Steps	Expected Result	Priority
			missing SWIFT Code			SWIFT Code 2. Save Account	mandatory only under International toggle (FR-904)	
TC-EXC-06	M-09	GET /api/bank-accounts	Quick Select fetch fails (network blip)	Simulate /api/bank-accounts timeout	Network timeout	1. Open Quick Select dropdown while the endpoint times out	Dropdown shows an empty/error state rather than hanging indefinitely; manual bank entry remains available as a fallback	Low

10.3 Exception Suite

TC ID	Module	API	Test Scenario	Preconditions	Test Data	Steps	Expected Result	Priority
TC-EXC-01	M-07	POST /api/invoices	MongoDB unreachable during save	Simulate DB connection failure (staging)	Valid complete invoice payload	1. Disconnect MongoDB 2. Submit Create New Invoice	User-facing error (“Unable to save invoice, please try again”); no partial document written; retry succeeds once DB is restored	High
TC-EXC-02	M-08	Mail dispatch endpoint (not documented)	SMTP/Brevo unavailable	Simulate SMTP outage	Valid recipient list	1. Trigger Mail send while SMTP is down	Dispatch fails gracefully with a clear error; invoice record unaffected; no duplicate emails on retry	High
TC-EXC-03	M-08	PDF download endpoint (not documented)	PDF generation throws an error	Simulate PDF engine failure (malformed Japanese font asset)	Corrupted/missing font asset	1. Attempt Download in Japanese while the katakana converter fails	Download fails with a clear error rather than delivering a corrupted/blank PDF; English download remains unaffected	Medium

11. Negative Testing

Negative cases derived from the source's own Negative Scenario suite (§4.2) and Business Rules. Items with no source coverage are marked accordingly rather than assumed.

- Missing mandatory fields — any field marked with a red * across all ten modules; each blocks submission with “This field is mandatory” (BR-04, FR-701, §7.1.3 item 1).
- Null / empty strings on required fields — covered for Company Address (TC-CLIENT-N01), IFSC Code (TC-BANK-N01), Bank Code (TC-BANK-N02), SWIFT Code under International (TC-BANKPAGE-N01).
- Invalid data types — not explicitly covered in the source (e.g. a string sent for a numeric Hours/Unit Price field); to be confirmed with development team.
- Invalid formats — email format validation is implied (RFC 5322 format required per Data Dictionary) but no explicit malformed-email test case is documented; to be confirmed with development team.
- Negative values — confirmed for tax rate (TC-TAX-E05: CGST = -5, must be rejected or clamped to 0, never produce a negative tax line).
- Zero values — confirmed for Hours = 0 (TC-CALC-E01, blocks the Add-Item gate) and tax rate = 0% (TC-TAX-E04, accepted).
- Extremely large values — confirmed for Hours/Unit Price (TC-CALC-E02: Hours=99999, Unit Price=999999, must compute without overflow/precision loss).
- Special / unsupported characters — covered specifically for Japanese character round-tripping (NFR-L01); no documented test for control characters, HTML, or script injection in text fields — to be confirmed with development team.
- Invalid IDs / non-existing IDs — not documented for /api/invoices/{id} or /api/bank-accounts/{id} (e.g. expected 404 behaviour); to be confirmed with development team.
- Duplicate records — not documented for Company/Employee/Sender duplicates; §7.1.1 item 3 flags duplicate 3-letter prefix collisions as a case to “verify and document actual behaviour,” implying it is not yet a settled rule.
- Invalid authentication — confirmed: missing/malformed JWT → 401 on every /api/* endpoint (NFR-S01, §7.1.4 item 1).
- Unauthorized access — role-based authorization is not confirmed in the source (see §2.4); only the sender-identity enforcement (NFR-S02, BR-05) is confirmed, and that is enforced regardless of caller identity, not as a role check.
- Invalid JSON / malformed request — not documented; to be confirmed with development team.
- Invalid currency — taxCountry accepts only INDIA | JAPAN; behaviour on an unsupported value is not documented.
- Invalid exchange rate — NOT APPLICABLE; no exchange-rate concept exists in this system.
- Invalid tax — confirmed: above-cap (TC-TAX-N01, TC-TAX-N02) and negative (TC-TAX-E05) values.
- Invalid discount — NOT APPLICABLE; no discount concept exists in this system.
- Invalid date — not explicitly covered (e.g. Due Date before Date); to be confirmed with development team.
- Invalid date range — the Invoices-list From/To date filter (FR-803) has no documented negative case (e.g. From > To); to be confirmed with development team.

12. Boundary Value Testing

Boundary conditions taken directly from the source's Edge Case Scenarios (§4.3) and Numbering/Tax checklists (§7.1.1, §7.1.2). Limits not stated in the source are marked as unconfirmed rather than invented.

Boundary	Source Reference	Expected Behaviour
CGST/SGST = 9 (India cap, inclusive)	TC-TAX-E01	Accepted as-is; no clamping event

Boundary	Source Reference	Expected Behaviour
CGST/SGST = 9.01 (just over cap)	TC-TAX-E02	Clamps to 9 — whether decimals are permitted at all is a product decision to verify
CGST/SGST = 0	TC-TAX-E04	Accepted; tax line = ₹0.00
CGST/SGST < 0 (e.g. -5)	TC-TAX-E05	Rejected or clamped to 0; must never produce a negative tax line
Consumption Tax = 10 (Japan cap, inclusive)	TC-TAX-E03	Accepted as-is; no clamping event
Hours = 0	TC-CALC-E01	Amount = 0.00; row does not satisfy "> 0" gating for + Add Item (BR-07)
Hours = 99999, Unit Price = 999999 (very large)	TC-CALC-E02	Amount computes without overflow/precision loss; totals panel formats with correct thousands separators
Hours = 7.5 (fractional)	TC-CALC-E03	Amount = 750.00; math engine accepts fractional hours
Recipient company name < 3 letters (e.g. "Ai")	TC-INV-GEN-E01	System must define a fallback (e.g. pad, or use full name) — exact fallback rule not documented; verify actual behaviour
Recipient company name starting with a digit/symbol	§7.1.1 item 2	Must resolve to a usable prefix without crashing generation
Maximum Services rows / maximum invoice line items	Not documented	No maximum is stated in the source — to be confirmed with development team
Maximum invoice number length / sequence rollover (e.g. INV-ACC-9999 → ?)	Not documented	To be confirmed with development team
Very long Japanese text in a description/address field	NFR-L01 (implied)	Must round-trip without truncation or mojibake; explicit max length not documented — to be confirmed with development team
Very long customer name	Not documented	To be confirmed with development team
Password length boundary (minimum 6 characters)	Data Dictionary §17.1	Passwords under 6 characters should be rejected — no explicit test case documents the exact boundary behaviour; to be confirmed

13. Database Validation

How QA should validate API operations against Cloud MongoDB. Flow:

API Request → Spring Boot Backend Processing → Cloud MongoDB → API Response

Module	Collection	Fields to Validate	Insert / Update / Delete Notes
M-01 Auth	Not documented — to be confirmed with development team	email uniqueness, hashed password (algorithm to be confirmed, §7.1.4 item 6)	Insert on signup; no update/delete flow documented for user accounts
M-02 Sender	sender collection (BR-11)	Company Name, Address, Logo, Signature, From Email	Upsert on first invoice save via +Others...; no explicit update/delete documented
M-03 Recipient (Company)	companies collection (§1.3, BR-11)	Company Name, Email, Address, Phone	Upsert on first invoice save via +Others...; deletion behaviour on cascading invoices not documented
M-03 Recipient (Employee)	dynamicClientEmployees collection (Glossary §17.6, BR-11)	Employee Name, Email, Address, Phone	Upsert on first invoice save via +Others...
M-07/M-08 Invoices	Not documented — to be confirmed with development team	All Data Dictionary fields (§17); invoiceStatus (ACTIVE | SOFT_DELETED)	Insert on Create (POST /api/invoices); update on Edit (PUT /api/invoices/{id}); DELETE must flip invoiceStatus to SOFT_DELETED, never remove the document (BR-09, NFR-S03)
M-09 Bank Accounts	bank_accounts collection (FR-905)	region, bankName, accountNumber, accountHolderName, branchName, ifscCode / bankCode+branchCode, swiftCode, accountType	Insert on Save Account; Edit/Delete controls documented but endpoint/collection-level behaviour not detailed — confirm cascading effect on Quick Select (BR-12)

Cross-Cutting Checks:

Encoding validation: for any field expected to carry Japanese text (bankName, branchName, addresses, service descriptions, company names), confirm the MongoDB-stored value is byte-identical UTF-8, not a transliterated or corrupted form (NFR-L01). Currency

Module	Collection	Fields to Validate	Insert / Update / Delete Notes
values: confirm cgstRate/sgstRate/consumptionTaxRate are never stored above their caps (NFR-S04) and that grandTotal stored equals the exact subtotal + tax + roundOff calculation (BR-08), not a rounded intermediate value.			

14. API Response Validation

The source documentation confirms very little about response envelope shape. This section documents what is confirmed and flags the rest.

Aspect	Status
HTTP status code	Only 401 (missing/expired JWT, NFR-S01/NFR-S06) is explicitly confirmed as a status code. Success codes for Create/Update/Delete/List are referenced only loosely (e.g. "200/201" in the source's own TC-PERSIST-01) — to be confirmed with development team.
Response headers	Not documented — to be confirmed with development team.
Response body / JSON structure	Confirmed field-level for invoice list rows (Invoice Number, Client Name & Email, Date Created, Amount — FR-703) and for the JWT token on login. Full envelope shape (bare array vs. paged object, wrapper keys) is not documented — to be confirmed with development team.
Required fields	Per Data Dictionary §17 — see §8 above for the full mandatory-field list.
Data types	Per Data Dictionary §17 (String, Decimal, Date, Enum) — no JSON Schema is published; type coercion behaviour on the wire is not documented.
Null handling	Not documented — to be confirmed with development team (e.g. whether poNumber is returned as null, empty string, or omitted when not entered).
Error messages	Only two literal UI-level messages are documented: "This field is mandatory" (client-side validation, BR-04) and "Unable to save invoice, please try again" (TC-EXC-01, MongoDB unreachable). Neither is confirmed as the literal API error-response body — to be confirmed with development team.
Error codes	Not documented — to be confirmed with development team.
Response time	NFR-P02: Invoices list should render in < 2 s for up to 1,000 invoices with server-side pagination/filtering. NFR-P03: PDF generation < 3 s (English/Japanese) or < 5 s (Bilingual).
Data consistency	Confirmed expectation: the Total Invoices counter (FR-801) must always equal the live MongoDB count of ACTIVE invoices; Grand Total must always equal Subtotal + Tax + Round Off (BR-08).

15. Error Handling

An error-handling matrix built from the source's Exception Scenarios (§4.4) and Availability NFRs (§6.2). Exact HTTP status codes and error response bodies are largely unconfirmed in the source — do not assume a specific JSON error shape until verified against the backend.

Error Scenario	Expected HTTP Code	Expected Error Response	Validation
Missing/malformed JWT on any /api/* call	401	Not documented — exact error message to be confirmed with development team.	Confirm every protected endpoint rejects consistently (NFR-S01).
Expired JWT	401	Not documented — exact error message to be confirmed with development team. UI shows a "session-expired" message and redirects to Login.	Confirm the redirect does not silently drop unsaved form data without warning (TC-EXC-04).
CGST/SGST or Consumption Tax submitted above cap via raw API call	Not documented — to be confirmed	Not documented; the requirement (NFR-S04) is that the value is "rejected or clamped server-side; never persisted above the cap" — which of the two behaviours is implemented is unconfirmed.	Verify server-side behaviour independent of the UI clamp; do not rely on client-side clamping alone.
Mandatory field missing on POST /api/invoices	Not documented — to be confirmed (400 is the reasonable default)	Client-side message is "This field is mandatory"; server-side response body/message is not documented.	Confirm the server enforces BR-04 independently of the client-side guard (defense in depth).
Direct DELETE /api/invoices/{id} bypassing the confirmation modal	Not documented	Not documented.	Confirm the delete is a soft-delete (or is rejected), never a hard delete, per BR-09/NFR-S03.
MongoDB unreachable during Create Invoice	Not documented	"Unable to save invoice, please try again" (user-facing, TC-EXC-01) — whether this is the literal API error body or a UI-composed message is not documented.	Confirm no partial document is written; confirm retry succeeds once the DB is restored.
SMTP/Brevo unavailable during Mail dispatch	Not documented	Not documented; must be "a clear error" (TC-EXC-02).	Confirm the invoice is never marked "sent" on a failed send, and no duplicate email is sent on retry.
PDF generation failure (e.g. malformed Japanese font asset)	Not documented	Not documented; must be "a clear error rather than delivering a corrupted/blank PDF" (TC-EXC-03).	Confirm the underlying invoice record is not corrupted and other-language downloads remain unaffected (NFR-A03).
GET /api/bank-accounts timeout	Not documented	Not documented; UI must show "an empty/error state rather than hanging indefinitely" (TC-EXC-06).	Confirm manual bank entry remains available as a fallback.

16. Security API Testing

Security tests limited to what the source Access Security checklist (NFR-S01–S08, §6.1) and JWT/SMTP checklists (§7.1.4, §7.1.5) actually claim. Role-based access control is explicitly NOT claimed by the source and is therefore not tested here as if it existed.

- Authentication — all /api/* endpoints require a valid JWT Bearer token; requests without one return HTTP 401 (NFR-S01). Confirm this against invoices, companies, employees, bank-accounts, and mail-dispatch endpoints specifically (§7.1.4 item 1).
- Authorization (sender identity) — the outbound email sender is enforced server-side (BrevoEmailService.resolveSenderEmail()) and can never be overridden by a client-supplied “From Email” value; the invoice's displayed From Email is cosmetic only (NFR-S02, BR-05, TC-MAIL-N01).
- Token validation — confirm the JWT payload does not contain the plaintext password or any bank/SWIFT/IFSC data (§7.1.4 item 3).
- Access control — role-based authorization (Admin/Finance/HR/Recipient) is NOT confirmed anywhere in the source; do not test for role-gated 403 responses unless and until the development team confirms a role model exists (see §2.4, §25).
- Unauthorized API access — covered under Authentication above; no separate “wrong role” test exists in the source.
- Invalid token — a tampered/invalid JWT should be rejected the same as a missing one (401), per the general NFR-S01 statement; no dedicated tampered-token test case is documented in the source — recommend adding one, matching the JWT security pattern used elsewhere in comparable systems.
- Expired token — confirmed: TC-EXC-04, HTTP 401 on expiry, TTL value itself not documented (NFR-S06).
- Input validation — server-side tax-cap enforcement is explicitly required in addition to the client-side clamp (NFR-S04); no other server-side input-validation guarantee is documented.
- Injection-related validation — not documented; no SQL/NoSQL injection or XSS test case exists in the source. Given MongoDB is used, recommend adding NoSQL-injection-shaped negative tests as a QA-added enhancement, flagged here rather than presented as a documented requirement.
- Sensitive data exposure — confirmed: the Brevo/SMTP App Password must never appear in the frontend bundle, browser console log, or any API response body (NFR-S07, §7.1.5 item 1). Confirm no bank/SWIFT/IFSC data or plaintext password ever appears in a JWT or error response.
- HTTP method validation — not documented (e.g. whether an unsupported verb on /api/invoices/{id} returns 405); to be confirmed with development team.

17. Data Dictionary

Every field entering or leaving the Invoice System, reproduced from the source's Data Dictionary (§5) and grouped identically — input fields, output/computed fields, and status/enumeration values — for direct use in API request/response assertions.

17.1 Input Fields

Field	Module	Type	Mandatory	Format	Validation	Description
fullName	M-01	String	Yes (signup)	Free text	—	User's legal name captured at sign-up (FR-101).
email	M-01/M-03	String	Yes	RFC 5322	Must contain @ and domain	Login identifier and/or recipient email; validated for format.
password / confirmPassword	M-01	String	Yes (signup)	Min. 6 characters	Must match exactly	Never stored/transmitted in plaintext beyond the request (FR-101).
senderCompany	M-02	Enum/String	Yes	Vision AI | Ideal Folks | VCAS Consulting | custom	—	Drives autofill (FR-202) and prefix (FR-402); custom values persist to sender collection (FR-203).
fromEmail	M-02	String	Yes	RFC 5322	—	Cosmetic display only; SMTP dispatch always uses hr@visionai.jp (BR-05).
clientType	M-03	Enum	Yes	Company | Employee	—	Defaults to Company (FR-301).
recipientCompany / recipientEmployee	M-03	String/Object	Yes	Selected or manually entered	—	Drives invoice-number prefix (FR-401) and bank autofill source (FR-507).
recipientAddress	M-03	String	Yes	Multi-line free text	—	Autofilled or entered via +Others....
recipientEmail	M-03	String	Yes	RFC 5322	—	Used on the printed invoice and as the default Mail-dispatch recipient.
companyPhone / employeePhone	M-03	String	Yes (when +Others...)	Free text (numeric preferred)	—	Mandatory on manual entry (FR-302, FR-303).
invoiceNumber	M-04	String	System-generated	INV-[PREFIX]-[n]	Read-only	3-letter prefix or hardcoded prefix (FR-401, FR-402, BR-03).

Field	Module	Type	Mandatory	Format	Validation	Description
poNumber	M-04	String	No (poNumber?)	Free text	—	Strictly optional (FR-403).
date	M-04	Date	Yes	YYYY-MM-DD	—	Defaults to today (FR-404).
dueDate	M-04	Date	Yes	YYYY-MM-DD	—	Defaults to date + 45 days (FR-404).
taxCountry	M-05	Enum	Yes	India | Japan	—	Drives currency, tax fields, bank mandatory-field set (FR-501–506).
cgstRate / sgstRate	M-05	Decimal	Yes (India)	0–9 (%)	Hard-capped at 9	Default 9 each (FR-502, BR-01).
consumptionTaxRate	M-05	Decimal	Yes (Japan)	0–10 (%)	Hard-capped at 10	Default 10 (FR-504, BR-02).
bankName	M-05/M-09	String	Yes	Free text (Kanji/Katakana supported)	—	All regimes (FR-1001).
accountNumber	M-05/M-09	String	Yes	Free text (numeric preferred)	—	All regimes.
accountHolderName	M-05/M-09	String	Yes	Free text	—	All regimes.
ifscCode	M-05/M-09	String	Yes (India only)	11-char IFSC format expected	—	Not shown/required under Japan or International-only (BR-06).
bankCode	M-05/M-09	String	Yes (Japan, International)	4-digit numeric	—	Zengin System bank code (FR-506, FR-904).
branchCode	M-05/M-09	String	Yes (Japan, International)	Free text / numeric	—	Zengin System branch code.
branchName	M-05/M-09	String	Yes	Free text (Kanji supported)	—	All regimes.
swiftCode	M-05/M-09	String	No (India/Japan) / Yes (International)	8 or 11-char SWIFT/BIC	—	Mandatory only under International (FR-904, BR-06).
accountType	M-05/M-09	Enum	Yes	Savings | Current | ...	—	All regimes.
serviceDescription	M-06	String	Yes (row 1)	Free text	—	Feeds email attachment filename convention (BR-10).

Field	Module	Type	Mandatory	Format	Validation	Description
hours	M-06	Decimal	Yes (row 1)	≥ 0, default 0	Must be > 0 for Add-Item gate	FR-601, BR-07.
unitPrice	M-06	Decimal	Yes (row 1)	≥ 0, default 0.00	Must be > 0 for Add-Item gate	FR-601, BR-07.
primaryRecipientEmail	M-08 (Mail)	String	System-derived	RFC 5322	Read-only	Fixed to invoice's Bill To email; not editable in Mail modal (FR-809).
additionalRecipients	M-08 (Mail)	String	No	Comma-separated email list	Parsed on commas	Optional extra addresses (FR-809).
downloadLanguage / mailLanguage	M-08	Enum	Yes	English | Japanese | Both	—	“Both” yields two separate single-language files (BR-10a).
searchQuery	M-08	String	No	Free text	—	Filters by Invoice Number, Employee/Company Name, or Email (FR-802).
fromDate / toDate	M-08	Date	No	YYYY-MM-DD	—	Filters Invoices list to a created-date window (FR-803).

17.2 Output / Computed Fields

Field	Type	Format	Description / Validation Rule
amount (row)	Decimal	Currency, 2dp	Math.round(hours × unitPrice × 100) / 100; read-only (FR-602).
subtotal	Decimal	Currency, 2dp	Sum of every row amount (FR-603).
taxAmount (CGST / SGST / Consumption Tax)	Decimal	Currency, 2dp	subtotal × (taxRate / 100), per applicable component (FR-603).
roundOff	Decimal (signed)	Currency, 2dp	Adjustment (+/-) required to make grandTotal a whole integer (FR-603, BR-08).
grandTotal	Decimal	Currency, whole integer	subtotal + taxAmount(s) + roundOff (FR-603, BR-08).
currencySymbol	Enum	₹ | ¥	Derived from taxCountry (FR-501, FR-503).
dateCreated	Timestamp	ISO 8601	Set on first successful POST to /api/invoices (FR-703).
attachmentFilename (single-language)	String	[Invoice Number] [Description].pdf	Composed server-side (BR-10).
attachmentFilename (Both)	String	[Invoice Number] [Service Description] (English/Japanese).pdf	Final standard per Commit 0747c74 (BR-10a); used identically for Download and Mail.

Field	Type	Format	Description / Validation Rule
emailSubject	String	Invoice #[Invoice Number] - [Sender Company]	Composed via String.format() in BrevoEmailService.java; falls back to “Your Invoice” when Company Name is blank.
totalInvoicesCount	Integer	Whole number	Live count of persisted (non-deleted) invoices (FR-801).

17.3 Status & Enumeration Values

Field	Allowed Value	Meaning
taxCountry	INDIA	CGST + SGST regime; ₹ currency; IFSC-based bank fields.
taxCountry	JAPAN	Consumption Tax regime; ¥ currency; Bank Code / Branch Code bank fields.
bankRegime (Bank Accounts page only)	INTERNATIONAL	Adds mandatory SWIFT Code on top of the Japan-style field set (FR-904).
clientType	COMPANY	Default recipient type; B2B invoicing flow.
clientType	EMPLOYEE	Bank autofill sources from the employee's profile (FR-507).
senderCompany	VISION_AI | IDEAL_FOLKS | VCAS_CONSULTING	Hardcoded entities; trigger autofill + fixed invoice-number prefix.
senderCompany	CUSTOM	Any sender added via +Others...; persisted on first save (FR-203).
downloadLanguage / mailLanguage	ENGLISH | JAPANESE | BOTH	Selects PDF render language (FR-807, FR-809).
accountType	SAVINGS | CURRENT (dropdown-defined set)	Bank account classification.
invoiceStatus (implicit)	ACTIVE | SOFT_DELETED	SOFT_DELETED invoices are hidden but remain in MongoDB for audit (BR-09).

Source Note:

Numbering note: the invoiceNumber sequence is scoped per prefix, not global. INV-ACC-1, INV-ACC-2... increments independently from INV-VI-1, INV-VI-2..., and independently again from INV-IF- and INV-VCAS- sequences. A defect where two different prefixes share one global counter should be logged against FR-401/BR-03.

18. Test Data

Suitable test data categories, built only from fields and examples actually present in the source documentation. No synthetic PII beyond the source's own seeded demo data is introduced.

18.1 Valid Data

- Seeded demo environment (source §0.1): 3 registered sender entities (Vision AI, Ideal Folks, VCAS Consulting), sample company clients (KK Blue Arbarao, Accenture), one saved bank account (Andhra Bank — 1234567), and one saved GMO Aozora Net Bank (Japan) sender bank.
- Valid India invoice: taxCountry=INDIA, cgstRate=9, sgstRate=9, ifscCode present, all India-mandatory bank fields present.
- Valid Japan invoice: taxCountry=JAPAN, consumptionTaxRate=10, bankCode/branchCode present, ifscCode absent.

18.2 Invalid Data

- CGST/SGST = 15 or 20 (above the 9% cap) — TC-TAX-N01/N02.
- CGST = -5 (negative) — TC-TAX-E05.
- Empty IFSC Code under India / empty Bank Code under Japan / empty SWIFT Code under International — TC-BANK-N01, TC-BANK-N02, TC-BANKPAGE-N01.
- Empty Company Address on a +Others... recipient — TC-CLIENT-N01.

18.3 Boundary Data

- CGST/SGST = 9 exactly; Consumption Tax = 10 exactly (TC-TAX-E01, TC-TAX-E03).
- Hours = 0; Hours = 99999 with Unit Price = 999999; Hours = 7.5 (TC-CALC-E01–E03).
- Recipient company name = “Ai” (2 letters, under the 3-letter prefix rule) — TC-INV-GEN-E01.

18.4 Japanese Data

- Customer name: 山田太郎 (source's own IMPORTANT TEST example, §0).
- Company header convention: 合同会社 Vision AI (FR-1001, §7.1.6 item 6).
- Seeded Japan sender bank: GMO Aozora Net Bank (source §0.1).

18.5 English Data

- Seeded company client: Accenture, KK Blue Arbarao (source §0.1, TC-INV-GEN-01).
- Hardcoded senders: Vision AI LLC (JPY), Ideal Folks LLC (JPY), VCAS Consulting LLC (JPY).

18.6 Mixed-Language Data

- A description or address field combining English and Japanese, to verify addSpaceBetweenJaAndEn spacing enforcement (NFR-L02).

18.7 Tax-Regime Data

- India: CGST=9, SGST=9 (₹). Japan: Consumption Tax=10 (¥). There is no cross-regime or FX-conversion data set to prepare, since conversion is out of scope.

18.8 Discount Data

NOT APPLICABLE — the Invoice System has no discount concept.

19. Traceability

Maps Business Requirement / Rule → Module → API → Test Scenario → Test Case, using only IDs that exist in the source documentation. No requirement ID has been invented.

Requirement ID	Requirement	Module	API ID(s)	Scenario ID(s)	Test Case ID(s)
BR-01/FR-502	India CGST/SGST hard-capped at 9% each	M-05 Taxation	API-003	SC-09, SC-11, SC-12, SC-13	TC-TAX-N01, TC-TAX-E01, TC-TAX-E04, TC-TAX-E05
BR-02/FR-504	Japan Consumption Tax hard-capped at 10%	M-05 Taxation	API-003	SC-10, SC-11	TC-TAX-N02, TC-TAX-E03
BR-03/FR-401/FR-402	Auto-generated, sequential, per-prefix invoice numbering	M-04 Numbering	API-003	SC-04, SC-05, SC-06, SC-07, SC-08	TC-INV-GEN-01, TC-INV-GEN-02, TC-INV-GEN-E01
BR-04/FR-701	Mandatory-field gate blocks incomplete submission	M-07 Validation	API-003, API-006	SC-17, SC-20	TC-CLIENT-N01, TC-VALID-N01
BR-05/FR-810	Outbound email always originates from hr@visionai.jp	M-08 Mail	API-012	SC-23, SC-25	TC-MAIL-N01, TC-MAIL-N03
BR-06/FR-505/FR-506/FR-902–904	Per-regime mandatory bank fields (India/Japan/International)	M-05/M-09	API-003, API-009	SC-14, SC-15, SC-16	TC-BANK-N01, TC-BANK-N02, TC-BANKPAGE-N01
BR-07/FR-601	Services row-1 gating for + Add Item	M-06 Calculation	API-003	SC-18, SC-19	TC-SVC-N01, TC-CALC-E01–E03
BR-08/FR-603	Grand Total = Subtotal + Tax + Round Off (math engine authoritative)	M-06 Calculation	API-003	SC-19	TC-CALC-E01–E03
BR-09/NFR-S03	Invoices may only be soft-deleted; no hard-delete endpoint	M-08 Dashboard	API-007	SC-22	TC-DASH-N01
BR-10/BR-10a/FR-807/FR-809	PDF attachment filename convention (single and “Both”)	M-08 Dashboard/Mail	API-012, API-013	SC-24	TC-MAIL-N02
BR-11/FR-203/FR-302/FR-303	Custom sender/company/employee persisted on first save	M-02/M-03	API-015, API-016, API-017	SC-17	TC-CLIENT-N01
BR-12/FR-508/FR-905	Bank Accounts sync live to Quick Select	M-09 Bank Accounts	API-008, API-009	SC-26, SC-30	§3.3.18, TC-EXC-06
NFR-S01/FR-103	All /api/* endpoints require a valid JWT	M-01 Auth	All	SC-02, SC-03	TC-AUTH-N02, TC-EXC-04
NFR-S02/BR-05	Sender identity enforced server-side, not client-derived	M-08 Mail	API-012	SC-23	TC-MAIL-N01
FR-702	POST /api/invoices persists to Cloud MongoDB	M-07 Persistence	API-003	SC-21	TC-PERSIST-01
FR-801/FR-802/FR-803	Invoices list count, search, and date-range filter	M-08 Dashboard	API-004	—	— (no dedicated source TC beyond Dashboard

Requirement ID	Requirement	Module	API ID(s)	Scenario ID(s)	Test Case ID(s)
					screen reference §3.3.19)

20. End-to-End API Workflows

The canonical happy-path workflow, taken directly from the source's Business Process Overview (§2.1) and mapped onto the confirmed/inferred API calls from §6–§7.

Step	Action	API	Assertion
1	Sign up / log in	API-001, API-002	JWT issued; user lands on Dashboard
2	Select Tax Country (India or Japan)	Client-side only (no dedicated endpoint documented)	Currency symbol and tax fields switch across the form
3	Select From (Sender Company): hardcoded or +Others...	API-015 (not documented, custom sender only)	Hardcoded entity autofills Logo/Signature/Address/Email; custom sender opens manual entry, persisted on first save
4	Select To (Client): Company or Employee	API-016 / API-017 (not documented, custom recipient only)	Saved entry autofills Recipient Address + Email; +Others... persists on first save
5	Invoice Number & Dates auto-populate	Computed as part of API-003 payload / response	Invoice Number follows the sender's prefix rule; Date/Due Date default correctly
6	Bank Details autofill (or Quick Select override)	API-008 (GET /api/bank-accounts) for Quick Select	Correct source (sender vs employee) autofills; Quick Select overrides all bank fields
7	Complete Services row(s); totals recompute	Client-side calculation engine (no dedicated endpoint)	Amount/Subtotal/Tax/Round Off/Grand Total recompute per BR-08
8	Create New Invoice	API-003 (POST /api/invoices)	Mandatory-field validation passes; invoice persisted; redirected to Invoices page
9	New invoice appears at top of Invoices list	API-004 (GET /api/invoices)	Invoice Number, Client, Date Created, Amount visible immediately; Total counter increments (FR-703, FR-801)
10	Edit the invoice (Pencil icon)	API-005 (inferred GET), API-006 (PUT)	Mandatory-field validation re-applied identically to Create; change reflected on the row and the PDF
11	Preview the invoice (Eye icon)	API-014 (not documented)	Interactive modal renders the PDF without triggering a download
12	Download the invoice (English / Japanese / Both)	API-013 (not documented)	Correct filename convention (BR-10, BR-10a); “Both” yields two files
13	Mail the invoice	API-012 (not documented)	Envelope “From” is always hr@visionai.jp (BR-05); fixed body template (FR-810); correct attachment(s)
14	Soft-delete the invoice (Trash icon)	API-007 (DELETE /api/invoices/{id})	Record hidden from the active list but remains recoverable in MongoDB (BR-09)

21. Non-Functional API Testing

Taken directly from the source's Non-Functional Requirements (§6).

Area	Requirement	Target / Measure
Performance — Form	Calculation engine recomputes on keystroke without perceptible lag (NFR-P01)	< 100 ms recompute on a 20-row Services table
Performance — List	Invoices page loads and renders promptly as the dataset grows (NFR-P02)	< 2 s initial render for up to 1,000 invoices with server-side pagination/filtering
Performance — PDF	Single-invoice PDF generation completes within an acceptable window (NFR-P03)	< 3 s English/Japanese; < 5 s Bilingual
Reliability — Uptime	Available during core business hours (NFR-A01)	99.5% uptime over a rolling 30-day window, 08:00–20:00 JST/IST
Reliability — Mail degradation	SMTP/Brevo outage must not corrupt or block invoice save (NFR-A02)	Simulated SMTP outage: invoice save still succeeds; only Mail action fails with a clear error (TC-EXC-02)
Reliability — PDF degradation	PDF generation failure must not corrupt the invoice record (NFR-A03)	Simulated PDF failure: download fails cleanly; invoice record and other-language downloads unaffected (TC-EXC-03)
Data Durability	Every successfully created invoice is durably persisted (NFR-A04)	Backup/restore drill recovers 100% of test invoices created in the prior 24 hours
Character Encoding	UTF-8 Japanese round-trips correctly form → MongoDB → PDF (NFR-L01)	Zero mojibake/replacement-character defects across a Japanese-data regression pass
JA/EN Spacing	addSpaceBetweenJaAndEn enforces professional spacing (NFR-L02)	Visual regression check on Japanese and Bilingual PDF samples
i18n Coverage	Every UI label has an English and Japanese translation key (NFR-L03)	i18n key-coverage lint returns 0 missing keys per locale per release
Concurrency — Numbering	Concurrent invoice creation for the same prefix never yields a duplicate number (NFR-S08)	Load test: N concurrent creates for the same prefix yield N unique sequential numbers

Scope Note:

Load/stress/performance benchmarking itself is explicitly out of scope for this document (source §1.1, “Out of scope: Load / stress testing and performance benchmarking”); the targets above are recorded for reference and for QA sign-off tracking, not as a load-test plan.

22. QA Execution Checklist

A practical sign-off checklist, mirroring the source's own mandatory-sections gate (§0: "a pull request or sprint ticket must not proceed to development sign-off until this document passes QA review on all mandatory sections, with every functional requirement traced to at least one test case").

- ☐ Authentication verified (signup, login, JWT issuance — API-001, API-002)
- ☐ JWT enforcement verified on every /api/* endpoint exercised (401 on missing/expired token — NFR-S01, NFR-S06)
- ☐ All confirmed endpoints tested (API-001 through API-011)
- ☐ All "not documented" endpoints (API-012–API-018) confirmed with development team before being marked complete or incomplete
- ☐ Positive scenarios tested for every module (§9)
- ☐ Negative scenarios tested (§11)
- ☐ Boundary/edge scenarios tested (§12)
- ☐ Exception scenarios tested — MongoDB, SMTP, PDF engine, Bank Accounts fetch (§10.3)
- ☐ Database validation completed for Sender, Companies, Employees, Bank Accounts collections (§13)
- ☐ India CGST/SGST cap (9% each) verified client- and server-side (BR-01, NFR-S04)
- ☐ Japan Consumption Tax cap (10%) verified client- and server-side (BR-02, NFR-S04)
- ☐ Grand Total = Subtotal + Tax + Round Off verified across ≥ 10 varied Hours × Unit Price combinations (BR-08, §7.1.2 item 4)
- ☐ Per-regime mandatory bank fields verified (India / Japan / International — BR-06)
- ☐ English text verified across UI and PDF
- ☐ Japanese text verified end-to-end: FORM → API → DATABASE → API RESPONSE → PDF (§0, NFR-L01)
- ☐ UTF-8 encoding verified — zero mojibake across a Japanese-data regression pass
- ☐ PDF output verified for all three language modes (English / Japanese / Both), including the two-file behaviour of Both (BR-10a)
- ☐ Attachment filename convention verified for Download and Mail, all language modes (BR-10, BR-10a)
- ☐ Outbound sender identity verified as hr@visionai.jp regardless of invoice From Email (BR-05, NFR-S02)
- ☐ Soft-delete behaviour verified — no hard delete reachable via the API (BR-09, NFR-S03)
- ☐ Error handling verified against §15, with every "not documented" item escalated to development rather than assumed
- ☐ Regression testing completed against the Numbering, Tax/Rounding, and Mandatory-Field checklists (§7.1.1–§7.1.3 of the source)

23. Defect Reporting Guidelines

Recommended defect format, aligned with the source's own "How to record results" guidance (§4 intro) and Resolution Log conventions (§7.3).

Field	Guidance
Defect ID	Team defect-tracker ID.
Module	One of the 10 modules (§4) — e.g. M-05 Global Taxation & Bank Autofill.

Field	Guidance
API ID / Endpoint	From §6 Endpoint Inventory (e.g. API-003, POST /api/invoices), or “not documented — endpoint path unconfirmed” if applicable.
FR / BR Violated	The specific FR-ID or BR-ID from the source (e.g. BR-01, FR-502) — every defect should trace to one.
Environment	Dev / Staging / Production; build or commit hash if available.
Tax Regime / Sender-Recipient Combination	India or Japan; which sender and recipient combination was used — required per the source's own recording guidance (§4 intro).
Preconditions	State of the system before the test (e.g. “Invoice exists, Manager token available”).
Request	Full HTTP request — method, endpoint, headers (redact the JWT value itself in shared reports), body.
Test Data	Exact values used (e.g. CGST = 15).
Actual Result	What actually happened, with the raw response body/status.
Expected Result	What the source documentation (cite the FR/BR) says should happen.
HTTP Status	Actual vs. expected, noting explicitly when the expected code is itself not documented and was inferred.
Severity	Critical (data loss / soft-delete bypass / security bypass), High (business rule not enforced, e.g. tax cap bypass), Medium (wrong status code, missing validation), Low (message wording).
Priority	Team-assigned.
Screenshot / Logs	UI screenshot and/or raw API logs.
Database Evidence	MongoDB document before/after, especially for soft-delete, tax-cap, and Grand Total defects.

24. Final API Coverage Summary

Coverage tables to be completed during test execution. Cell values are left as — (not yet executed) rather than fabricated pass/fail numbers; populate during the QA run.

24.1 Module Coverage

Module	Total APIs	Tested	Passed	Failed	Blocked
M-01 Authentication	2	—	—	—	—
M-02 Sender Management	1 (not documented)	—	—	—	—
M-03 Recipient Management	2 (not documented)	—	—	—	—
M-04 Numbering & Dates	0 dedicated (part of API-003)	—	—	—	—
M-05 Taxation & Bank Rules	0 dedicated (part of API-003)	—	—	—	—
M-06 Calculation Engine	0 dedicated (part of API-003)	—	—	—	—
M-07 Validation & Persistence	1 (API-003)	—	—	—	—
M-08 Invoices Dashboard & Lifecycle	7 (API-004–007, 012–014)	—	—	—	—
M-09 Bank Accounts	4 (API-008–011)	—	—	—	—
M-10 Advanced Localization	1 (not documented)	—	—	—	—

24.2 Test Type Coverage

Test Type	Planned	Executed	Passed	Failed
Positive	31 scenarios (§9)	—	—	—
Negative	13 scenarios (§11 checklist)	—	—	—
Edge / Boundary	15 boundaries (§12)	—	—	—
Exception	6 scenarios (§10.3)	—	—	—
Security	11 checks (§16)	—	—	—

24.3 Requirement Coverage

Requirement Group	APIs Covered	Test Cases	Coverage Notes
BR-01 through BR-12 (12 rules)	3 APIs (API-003, API-009, API-012)	≥ 20 test cases across §10	Full traceability in §19; execution % pending QA run
FR-1xx through FR-10xx (documented FR set)	11 APIs confirmed + 7 named-but-undocumented	See §19	API-012–API-018 cannot reach 100% API-level coverage until endpoints are confirmed (§25)

25. Information Missing / To Be Confirmed with Development Team

Every API-relevant item that could not be determined from the provided Invoice System Functional & Test Documentation (v2.7). Nothing below has been invented — each line states exactly what is known and what is missing.

25.1 Environment & Architecture

- Base URL for Dev, Staging, and Production environments.
- Whether an API version segment (e.g. /api/v1/...) exists anywhere, or whether all routes sit directly under /api as the confirmed paths suggest.
- Exact Content-Type / Accept header requirements per endpoint.
- Whether a health-check endpoint exists.

25.2 Endpoints Not Explicitly Documented

- Exact REST path/verb for Mail Dispatch (API-012) — known to be backed by EmailController.java / apiService.ts.
- Exact REST path/verb for PDF Download (API-013) and PDF Preview (API-014).
- Exact REST path/verb for custom Sender persistence (API-015, sender collection).
- Exact REST path/verb for custom Company persistence (API-016, companies collection).
- Exact REST path/verb for custom Employee persistence (API-017, dynamicClientEmployees collection).
- Exact REST path/verb for Excel import/export (API-018, excelService.ts) — the source's own Resolution Log (§7.3) notes Module 10's screen-reference documentation is still pending as of v2.7, though development has confirmed the underlying engine is functional.
- Whether GET /api/invoices/{id} (API-005) exists as a standalone call, or whether Edit/Preview are served entirely from client-side state already loaded by GET /api/invoices.
- Exact REST path/verb for Edit (PUT) and Delete on a saved Bank Account (API-010/API-011) — only the UI controls are documented (FR-905).
- Query parameter names for the Invoices list search (FR-802) and From/To date filter (FR-803), and for any pagination on GET /api/invoices.
- Whether a Logout endpoint exists server-side, or whether logout is purely a client-side token clear (FR-103 only confirms the client-side effect).

25.3 Response Contracts

- Exact success HTTP status codes for Create/Update/Delete on Invoices and Bank Accounts (200 vs 201 vs 204 is inferred, not confirmed).
- Exact error response JSON shape for any endpoint — no error-body schema is published in the source.
- Full response envelope for GET /api/invoices and GET /api/bank-accounts (array vs. wrapped/paged object).
- Whether API responses declare charset=utf-8 explicitly for the Japanese-text guarantee in NFR-L01.

25.4 Business Logic Details

- Server-side behaviour when a raw API call submits a tax rate above the cap: rejected outright, or silently clamped (NFR-S04 requires one of the two but does not say which).
- Fallback rule when a sender/recipient company name is shorter than 3 letters for prefix generation (TC-INV-GEN-E01 flags this as unresolved).
- Behaviour when two different sender names collapse to the same 3-letter prefix (§7.1.1 item 3, explicitly flagged as “verify and document actual behaviour” in the source).
- Duplicate-detection rules (if any) for Companies, Employees, and Senders added via +Others....

- Maximum number of Services rows / line items per invoice, and behaviour at that limit.
- Maximum field lengths for free-text fields (customer name, address, description), especially for long Japanese text.
- Whether Company/Employee/Sender records can be updated or deleted via the API after first save (only creation/persistence is documented, BR-11).
- Role-based access control model — not confirmed anywhere in the current source; an earlier draft's assumed-roles table (Admin, Finance/Accounts, HR, Recipient) was explicitly removed pending dev/product confirmation (source §2, end-of-table note).
- Password-hashing algorithm at rest (BCrypt is suggested as an example in the source's own checklist item, not confirmed).
- JWT TTL (token lifetime) value.

25.5 Module 10 (Advanced Localization) Documentation Gap

Source Note:

Per the source's own Resolution Log (§7.3): screen coverage for Modules 1–9 is complete, but Module 10 (Excel import/export, a pure Japanese-language PDF sample) still lacks screen captures as of v2.7. Development has confirmed the underlying engines (pdfService.ts, backend ExcelService) are fully tested and functional, and describes this as a non-blocking documentation task to be completed during a final visual audit round. API-018 in this document should be revisited once that documentation lands.